

‘Everplane’ Vehicle Definition

- Group Design Project -



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Abstract

This paper forms part of the *Everplane* Group Design Project and aims to present the Vehicle Definition of an autonomous high-altitude long-endurance concept aircraft, designed for flying continuously over one month in a sustainable manner. The platform has the capability of acting as an aerosatellite, providing affordable internet to remote areas. Note that this report is part of 20 others approaching this design challenge. The focus points of this particular report are: initial and final aircraft configuration design, accurate mass and centre of gravity determination and computer-aided design for technical (CFD, FEA, Drawings) and media (rendering) purposes. The results, consistent with previous studies ^{[1] [2] [3]}, indicate that an optimum configuration is a conventional one, featuring a high aspect ratio (25) wing, a cruciform tail, 4 electric engines powered by hydrogen fuel cells, recharged by a large solar panel array. Furthermore, the total mass for the most recent aircraft iteration was determined as 539.65kg, within the flight allowance of 559kg. Finally, the concept was determined to be technically feasible from the perspective of all teams.

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List of Abbreviations

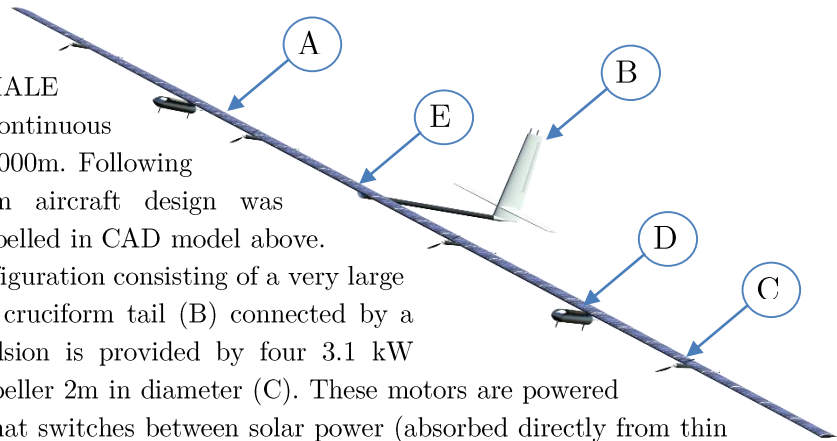
A/C	= Aircraft
AR	= Aspect Ratio
CAD	= Computer Aided Design
CFD	= Computational Fluid Dynamics
CG	= Centre of Gravity
FEA	= Finite Elements Analysis
GDP	= Group Design Project
HAA	= High Altitude Airship
HALE	= High Altitude Long Endurance
HTA	= Heavier-than-Air
LE	= Leading Edge
LTA	= Lighter-than-Air
MAC	= Medium Aerodynamic Chord
MARV	= Modular Automatic Runway Vehicle
Re	= Reynolds Number
T/O	= Take-Off
TE	= Trailing Edge
UAV	= Unmanned Aerial Vehicle

Executive Summary

Global Vehicle Layout

Everplane is defined as an autonomous, solar-powered, HALE aircraft, capable of 1 month continuous operation at a cruise altitude of 20,000m. Following an iterative process, an optimum aircraft design was determined. Its main features are labelled in CAD model above.

The aircraft is of a conventional configuration consisting of a very large aspect ratio main wing (A) with a cruciform tail (B) connected by a light-weight truss structure. Propulsion is provided by four 3.1 kW motors, each driving a 2-bladed propeller 2m in diameter (C). These motors are powered by an energy management system that switches between solar power (absorbed directly from thin silicon cells covering the upper surface of the main wing) and fuel cells located in two underwing pods (D). An additional pod mounted centrally under the main wing contains the avionics and payloads (E). The majority of Everplane's 20 kg payload is composed of antennae and transmitters to relay internet data. Six of these were mounted horizontally at the trailing edge of the main wing and six mounted vertically, integrated into the vertical tailplane and pod structures.



Alternative Configuration Studies

CAD and technical sketches were used throughout the design process to study alternative configurations. The final two contenders for the baseline configuration were the swept flying wing, and the conventional unswept wing with a tailplane. . Vehicle definition took an increasingly focused approach, from the overview (the baseline configuration), to engine placement and details such as optimum antennae positions. The position of all components was carefully chosen following a trade-off analysis between aerodynamic, structural and flight dynamics considerations. Our final configuration was chosen following an in depth study of performance and stability characteristics, making use of the CAD models for the purpose of FEA and CFD analysis.

Vehicle Mass & Centre of Gravity Analysis

The initial weight estimation was based on interpolation methods, empirical relations and relevant equipment available on the market. Two types of constraint diagrams were then utilized to determine key design parameters. The first of these took into consideration the amount of sunlight energy available, outputting an optimum wing loading which would result in the most compact aircraft possible. This was then applied to the classic power to weight ratio constraint diagram, subsequently a feasible design point was chosen ^[T3]. Subsequently, a new total aircraft weight of 559.4kg was estimated, based on further refinement of the configuration and more accurate information provided by other FATTs. This value was then used by all FATTs for the final iteration. A higher accuracy model of the mass and location of each part was developed, in order to determine the aircraft mass and the intervals where its CG could be placed. Through weight optimization and stability estimation, the final values presented below were obtained. Once the CG's of major components were established, using crude shape approximations, the mass moments of inertia were calculated.

Estimated Final Weight (kg)	Accurate Final Weight (kg)	CG Location:	X	Y	Z
559.4	539.65	From Leading Edge (m)	(pitch)	(roll)	(yaw)
		% of wing MAC	0	0.76	0.12
				34.96	

Table 1: Summary of final weights and global CG location

I_{xx} (kgm ²)	I_{yy} (kgm ²)	I_{zz} (kgm ²)
4610	151000	17400

Table 2: Mass moments of inertia

1. Introduction

The engineering challenges that need to be overcome in order to achieve the objectives set out by the Mission Definition Team ^[T1] for Everplane, make the concept design of this aircraft a particularly interesting problem, one that has received significant attention from both researchers and the public. However, from a vehicle definition perspective, the vast applications of this ‘aerosatellite’ platform are inconsequential, coming as a consequence of the aircraft being able to meet its requirements: climb safely through the troposphere within allowed time (30h), cruise at 20km for 1 month, carry a payload of 20kg, descend (within 7h) and land safely for a quick turnaround time. All with 0 greenhouse emissions. This has never been achieved in the past. Therefore, obtaining a viable vehicle definition for this task is of significant importance.

The design process involved the coordination of the Vehicle Definition Team with another 5 subgroups. Following the proposal of new vehicle configurations, the pro & against opinions of other subgroups (aerodynamics, structures, propulsion, stability and payloads), justified by numerical arguments, were taken into consideration to evolve the aircraft concept. Clear communication procedures were established to facilitate this process and as a result a number three design iterations with multiple alternative configuration studies one each of them, have been successfully completed. For this purpose a timeline with self-imposed deadlines (Annex A), similar to those used in industry ^[4] has been created.

2. Vehicle Alternatives Study

This chapter presents the evolution of the Everplane aircraft concept in a chronological manner, through 3 iterations, each with an average of 3 sub-iterations (relevant 2D concept sketches included in Annex B). This paper focuses on the initial and final configurations.

2.1 First Iteration

2.1.1 A historical perspective

Following a research of existing and concept A/C with similar mission requirements, a database was created with parameters relevant to the Everplane mission requirements. This selection was narrowed down to the best representative from each A/C type, with the use of a trade-off matrix analysis. Subsequently, a merit function (1) was formulated, to allocate points according to the level at which each A/C selected approached the desired requirements. Note that the coefficients of the merit function were approximated from the trade-off matrix analysis as 10, 8, 8 and 10 respectively. Table 1 summarises the results of this research and objective analysis.

$$M = C_1 \text{Endurance} + C_2 \text{Cruise Altitude} - 20 + C_3 \text{Payload} - 20 / \text{Empty Mass} + C_4 AR \quad (1)$$

Aircraft	Type	Configuration	Propulsion	Energy Storage	AR	Payload (kg)	Empty Mass	Altitude (km)	Endurance (days)	Merit Points
NASA Helios [5]	UAV	Flying Wing	14 brushless DC electric motors	Solar + H2 cells	31	329	599	32	1	420.13
Qinetiq Zephyr	UAV	Conventional	4 brushless DC electric motors	Solar + LiS Bat.	22	2.5	50.5	18	14.5	346.23
Solar Impulse	Pilot	Conventional	4 10 HP electric motors	Li-Ion Bat.	20	500	1500	13	1.5	158.56
Phantom Eye	UAV	Conventional	150 HP H2 Piston Engine	Liquid hydrogen	15	907	825	14	10	210.60
Voyager	Pilot	Canard twin-boom	130 HP Fossil Piston Engine	Fossil fuel	33	3000	1020.6	3.4	9	306.56
Lockheed HAA	UAV	Airship	2 kW electric motors	Li-ion batteries	N/A	22.7	300*	21	100+	1008.07

* Data not available to the public. **Best merit function.**

Table 3: Historical Aircraft Database final selection performance parameters and merit function point.

2.1.2 Baseline configuration

Although the Lockheed Martin High Altitude Airship received the highest number of merit points it was discarded as LTA vehicles are not the focus of this project. Therefore, the baseline configuration was narrowed down to the NASA Helios and Qinetiq Zephyr concepts. It is worth noting the fact that an endurance of 1 month worthy of the ‘aerosattelite’ designation has never been achieved in the past from a HTA aircraft. Helios experienced catastrophic structural failure upon ascent, due to turbulence. However, this is now well documented and has been taken into consideration [5] [6]. Zephyr’s energy storage system is not suitable for the number of recharge cycles required to reach 1 month endurance [7] and battery replacement implies high maintenance costs [8]. Due to the higher merit points, the first iteration baseline configuration was chosen based on the NASA Helios design, as a 75m flying wing with distributed propulsion systems. Stability and control considerations were to be analysed. Given the fact that there is no historical precedence for this aircraft, our team decided to opt for the best possible case for the baseline, to first determine it is feasible under best circumstances. Therefore, a powered landing gear independent of the A/C, was assumed (3.1.3), such that the wing structure would not be required to support its weight [T4] and only climb and cruise capability is expected from the propulsion systems [T6].

2.2 Second Iteration

The optimum conceptual design process recommended by Raymer [10] (schematics included in Annex C), and that adapted by Rizzo [11] to include particularities of a HALE A/C were followed throughout the alternative configurations study. To assure stability comparison between a swept wing and a conventional design yielded results in favour of the latter.

2.3 Final Iteration

The conceptual evolution of the A/C is graphically presented in Annex B, including a progress log in Annex X. This subchapter focuses of the refinement of the 3rd (final) iteration.

2.3.1 Overview

For the final iteration the wing dimensions and aircraft mass budget have been established. The wing has a 54.62m span with a straight central section measuring half the span and linear taper towards the tip, from a root chord of 2.43 to 1.46.

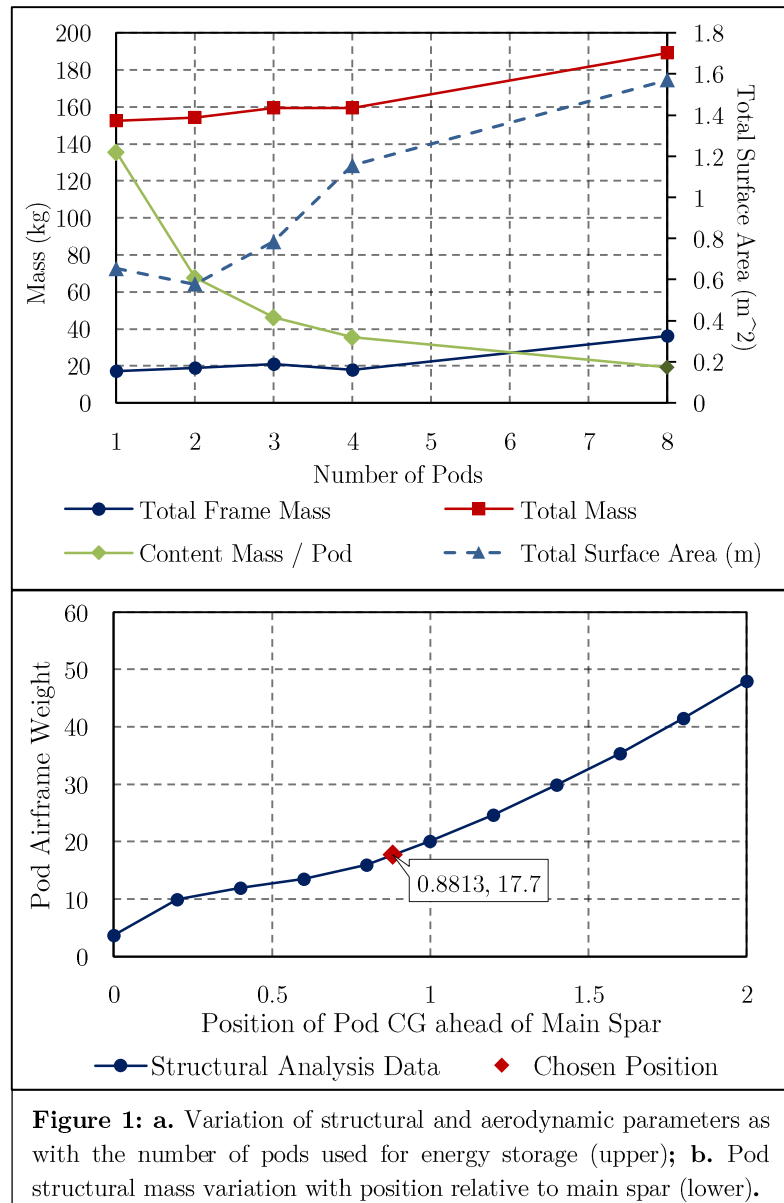
2.3.2 Tail Alternatives

Various tail designs were considered in collaboration with the following teams (in order of priority): Flight Dynamics, Aerodynamics, Structures. V tail was dismissed due to potential interaction with wake and control complexity. The use the fuselage as a yaw control surface, by placing it at an angle with respect to the chordline and shaping it as control surface was also dismissed due to structural mass considerations^[T4] and increased profile drag associated with this angle. The configuration was narrowed down to a T tail and a cross tail. Although the first had a lower induced drag estimate^[T3] it suffered a significant mass increase compared to second option. In the end, the cross tail was chosen as the best option.

2.3.3 Pods Placement

A trade-of analysis was performed in order to determine the optimum number of pods, as summarised by Figure 1a. Structural mass increase is not significant (within 10kg) between 1 and 4 pods, however there is a sharp increase in surface area and associated skin friction drag for 4 or more pods. Thus, a two pod design was chosen, for the power storage system. This enables an even distribution of the H₂ and O₂ storage, only one being needed for each chemical. Two pods also reduce the number of connectors required between tanks.

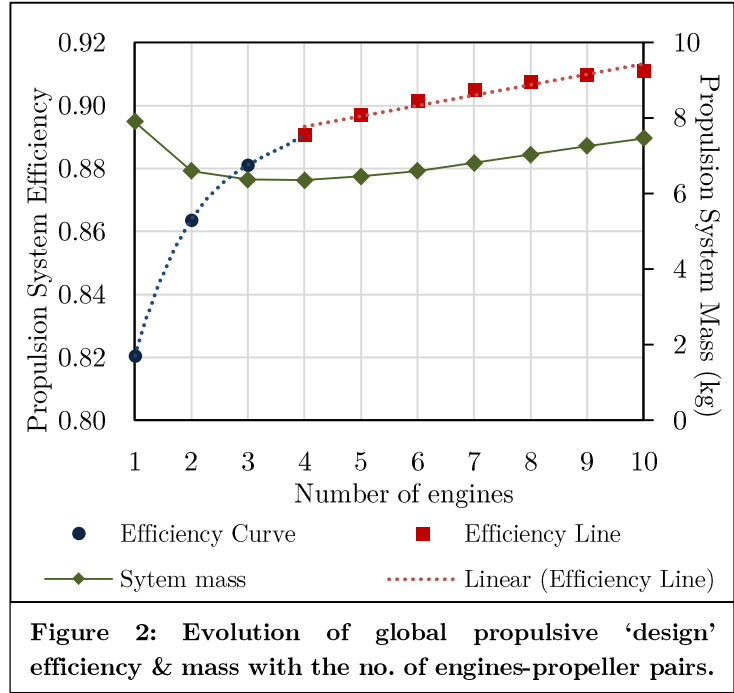
Chordwise placement affects wing-mounts structural mass. Moving the pod CG away from the load-bearing main-spar brings a weight penalty as shown in Figure 1b. However, pods location has a significant effect on global CG location. Therefore, a compromise was carried out, by the CG location script (4.3).



Spanwise placement is symmetric for an even number of pods. However, the exact position is largely affected by structural considerations. In order to prevent excessive wing-bending and reduce structural weight the pods location has been chosen as $\pm$ half-span ^[T4].

2.3.4 Engine Configuration

Ground clearance was set to 3 m in the previous iterations, implying a maximum propeller diameter of 2 m. Coincidentally, this blade size was found to provide close to the optimum efficiency for flight conditions ^[T6] at cruise. Typically a larger propeller diameter implies a higher efficiency per engine, however in the flight regime this was not the case. Because the total thrust requirement does not vary with engine number, a higher number of engines implies a lower engine RPM leading to a decrease in



Re. Because of poor research into low Re aerofoils. There are no significant efficiency gains beyond a certain point, as can be seen from Figure 2. At a number of 4 engines, the efficiency behaviour transitions from a curve into a line. This transition point was found to be optimum from a propulsion system mass point of view as well. The following assumptions were made:

- All propellers are of same diameter (2m) use the same number of blades (2).
- They employ the same aerofoil (MH 112), but blade geometry is optimised.
- No blade structural weight variation with decreasing RPM.
- Propulsive system mass was obtained from performance data interpolation, based on engines currently available on the market.
- Drag variation associated with the number of engines assumed negligible.
- All engines providing a thrust equal to the total thrust requirement/no. engines.

Considering all of the above, the optimum number of engines was established as 4.

The next stage of engine configuration analysis is placement. For this purpose, the use of distributed electric propulsion for the purpose of drag reduction was studied ^[11] and through aerodynamic analysis ^[T3] it was established that a total wing lift increase of 5% is correlated with the use of 4 engines. Therefore, there is a local lift increase over the wing section behind the propellers (which as located in front of the wing leading edge). This invites the idea of

placing the engines along the span such that the percentage increase in local lift behind them supports the larger mass concentrations of the fuel cell pods and hence reduces the loads paths through the structure and associated airframe mass, as per the figure below.

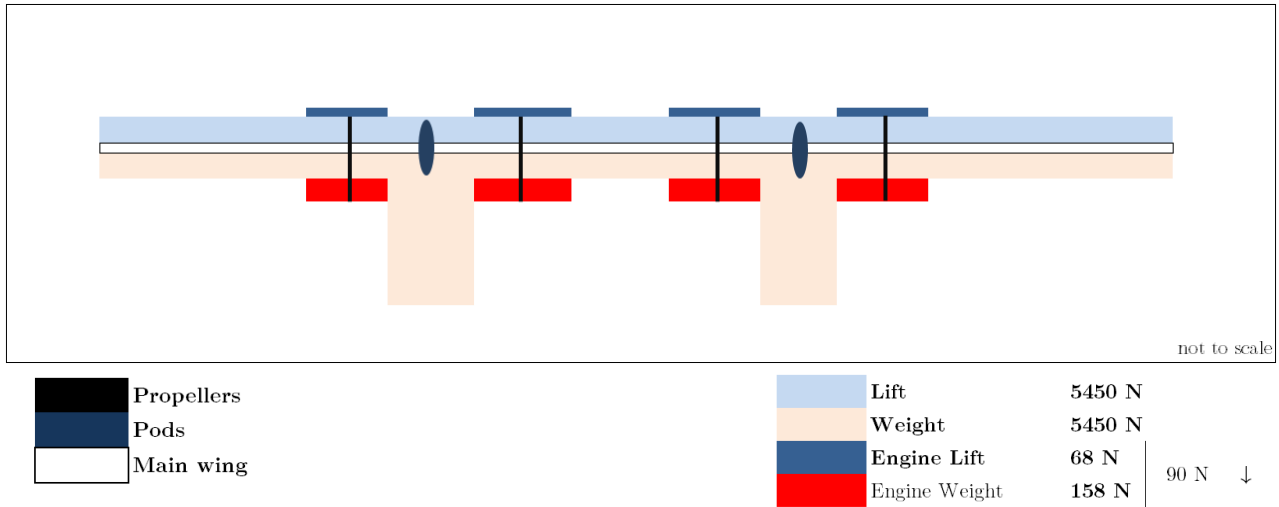


Figure 3: Infographic depicting vertical force distribution along the main wing and local lift increase.

The above analysis proved that the local lift generated was not sufficient to justify the engines being placed near the main pods, as the weight of the propulsion system is higher than the lift increase it provides. The 34% lift increase over the area in the propeller streamline and the engine weight imply a 90N downward force resultant behind each engine. Therefore, there is no advantage to placing the engines closer to the main pods, except the reduction in wire length. In fact, this engine placement would be a disadvantage in terms of structural weight because it increases the mass concentration at these positions. Furthermore, interference drag between the pods and the engines was also taken into account, but due to the low engine/pod diameter ratio, it was found to be insignificant (<1%) if the distance between the two was higher than 2.8m [10]. Placing the engines near the tips to alleviate wing bending was also considered, but due to taper, the structure is not capable of holding the engines at outward positions without additional reinforcements [T4]. Therefore, through a compromise of the above it was decided that the engines should be evenly distributed along the span, equidistant from the main pods (at $\pm 3s/4$ and $\pm s/4$) to ensure optimum structural and aerodynamic efficiency.

The chordwise position of the engines is driven by global A/C CG position considerations. This was chosen as 1 m in front of the wing LE as a compromise between favourable stability (shifting the engine forward lowered the CG with respect to the A/C pitch axis) [T7] and structural mass [T4] associated with the engine shaft and the frame onto which it is attached. As in the case of the main pod, an increase in the distance from the main spar (located at the position of maximum aerofoil thickness, 0.85m behind LE) results in an increase in mass.

3. Assisted Launch & Take-off

The decision to optimise the aircraft for cruise in order to obtain the best possible probability for designing an aircraft iteration capable of achieving mission requirements eliminated the possibility of a landing gear due to the associated structural mass increase. This chapter presents alternative configurations designed to enable safe insertion/extraction from cruise.

3.1 Take-off

3.1.1 Initial Analysis

All take-off options available in aerospace literature have been subjected to analysis to determine the applicability to the mission requirements and compatibility with the A/C fragile structure. Among the options analysed are water & grass skids, detachable gears, human-assisted, rail and catapult launchers. A ‘mothership’ assisted-launch (such as that used by Virgin Galactic) was considered. However, due to the lack of candidate planes capable of the low speeds associated with Everplane an airflow shield as well as a support structure would need to be retrofitted to the plane, and insertion to altitude would require operation under the stall speed of the ‘mothership’ at high altitude, which is not feasible. In the final steps of the analysis of the final iteration, the Propulsion Team observed a miscalculation which implied that Everplane could no longer satisfy climb requirements from ground altitude due to lack of available power. As a consequence Vehicle Definition analysed multiple alternatives to a self-climbing A/C to get to the desired altitude (Annex I). In the end, it was established that, although margins are small the A/C will self-climb.

3.1.2 Balloon Launch

This launch method has several advantages in addition to removing the need for a landing gear. It allows the bypassing the turbulent layer of the atmosphere (the reason behind the NASA Helios accidents [7]), therefore there is no need to design for gust loads, no energy is consumed to get up to altitude, ascent times are much lower and it is possible to launch before dusk. However, there is an additional mass increase associated with the connection mechanism between the balloon and the A/C, estimated to 10kg^[T4]. Furthermore, the initial acceleration of the system can destroy the vehicle. To avoid the need of structural reinforcements the balloon would use a liquid-gas H₂ converter to vaporise the required amount of H₂ to create a buoyancy sufficient for a low positive acceleration. The mass of this system is associated with the balloon and was approximated as 60kg^[T6]. Ultimately the feasibility of launch system largely depends on A/C mass. Beyond a certain value it becomes cost-prohibitive to use this launch method. This launch method can place the vehicle up to

an altitude of 80km, however there is an exponential increase in balloon mass with target altitude and thus significant associated costs increase (Figure 4).

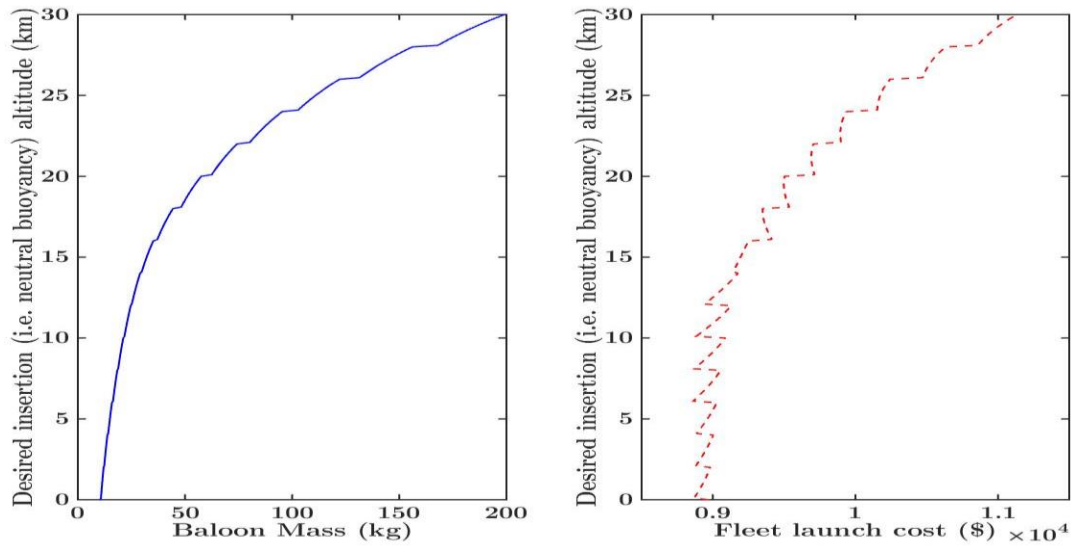


Figure 4: Output of the Matlab script, showing variation of launch cost with insertion altitude.

To determine the trade-off between launch costs and desired insertion altitude, a Matlab script was developed, based on data available from the Red Bull Stratos - High Altitude Balloon ^[13] project. H₂ cost, not including storage and transportation was approximated as 2.77 \$/kg ^[T1]. The script (included in Annex F) features nested loops that account for the increase in balloon size and envelope skin with neutral buoyancy altitude and payload mass. The output, revealed that for a target altitude of 20 km a balloon weighing 199.2 kg would be necessary (including: 15kg wires and quick release mechanism, 50kg vaporiser, 54.57±0.1kg of H₂, 10kg recovery parachute and 2kg control unit). Therefore, just the cost associated with launching 50 planes is 11,133 \$. Cost of the balloons, their recovery and the cost of storing H₂ also would need to be accounted for by [T1]. Following a comparison with online calculators ^{[14] [15]} that were checked against each other to assure validity, the required H₂ mass was found to be underestimated by the script by approximately 80% for the 20km altitude. This is due to two main assumptions: neglecting the system drag and acceleration and a linear variation of balloon mass with volume of envelope material used (full list of assumption in Annex F). However the trend observed in the script output (Figure 4) is accurate within the above approximations.

3.1.3 MARV

M.A.R.V. (Modular Autonomous Runway Vehicle) is a concept developed by the Vehicle Definition team specifically for the Everplane landing and take-off requirements. It is based on the autonomous concept-car system recently revealed by NASA. It is a drive-by wire vehicle featuring 4 ‘e-wheel’ engine-wheel units, capable of independently rotating 360 degrees. These units are completely self-contained and include cooling, spring and dampening systems. The main body of the vehicle holds batteries and control computers. MARV utilises 3 of these vehicles retrofitted to operate as a take-off and landing platform. The modifications include radar and laser distance tracking sensors on the sides, enabling the 3 platform to operate as a single entity, aligned within centimetres of their required position. As seen in Figure 5, sensors will be located to the **sides** of the vehicle and the **top** will allow the addition of a docking interface with the plane. Full technical drawing of each of the MARV individual units are included at the end of Annex H.



Figure 6: MARV System Platforms



Figure 5: E-Wheel Unit

The key to the system is the e-wheel unit. Following a preliminary Simulink model development of the system dynamics of a car featuring 4 e-wheels, it was apparent that from a control-accuracy point of view it is highly feasible to hold a three units at a predefined position relative to each other. The steady-state response is highly accurate and transients indicate that the units are within 22 cm of each other (assuming a 1ms response delay from each unit). Therefore, it is possible to control the MARV components as an entity even through accelerated movements. Knowing the T/W required for T/O as 0.0486 ^[T1] the minimum thrust required from the MARV was found to be within the parameters of a single e-wheel (0.386kN, with a 50% safety margin) ^[16]. The speed required for T/O and landing can also be achieved (1.3Stall_Speed=7.91 m/s), MARV units are capable of sustained speeds of 17.8 m/s^[16]. Furthermore, the system has the advantage of relieving the plane from any need to consume energy during take-off, as the engines do not need to provide the power required to accelerate to T/O speed, therefore propulsion systems only need to be optimised for climb and cruise. Finally, this concept is also a more sustainable option compared to the

balloon launch, as it has no carbon costs associated with processing the H₂ and the transportation of balloons for retrieval.

3.2 Landing Alternatives

Following the analysis carried out in the first weeks of the project it was established that the plane will not have a landing gear and is not capable of a ‘grass landing’ without damage to the propulsion systems and additional reinforcements to prevent airframe damage (that imply a higher total mass). The only viable alternative for landing without a gear is the novel MARV proposal. A preliminary study in the feasibility of such a system was carried with in the previous chapter with positive results.

This chapter continues the analysis of the MARV system, introduced above, with a focus on landing capabilities. A graphic showing the working principle of MARV system as a landing platform has been included in Annex E. Furthermore, Figure 7 presents the successive stages of this automated process: upon approach the aircraft receives information from an array of atmospheric sensors located on the approach path, enabling it compensate for low intensity cross-winds. The A/C turns off its engines and aligns all propellers with the wing LE, gliding towards to runway. The MARV system accelerates to match the A/C speed and using reflective markers on the underside of the A/C it aligns within 20cm of its underside. The A/C glides onto the MARV and touches it at the mid-span position and directly behind outer engines, holding its weight. The dampers and spring system prevented a bounce-off and a suction-mechanism (featuring multiple cavities within the landing surface terminated in silicon rings and connected to a vacuum system) locked the plane to the top of the MARVS upon contact. The A/C tail is supporting its own weight and a tail-strike is prevented as the landing system application point is behind the global CG location. The MARV system slowly brings the A/C to rest. Its 360° rotation capability allows for a quick taxi procedure and turnaround time.

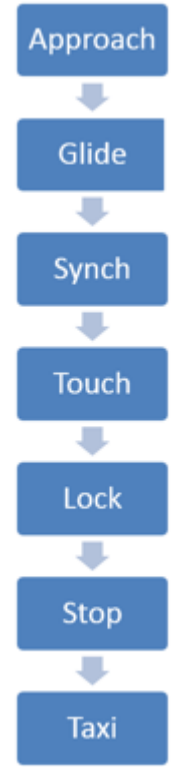


Figure 7: MARV Landing Procedure

An analysis of the minimum area required to prevent wing damage by the lock-in system was carried out to reveal that a surface of 0.2x0.2m / MARV is sufficient to prevent damage of the skin and ribs. The final design has a 2x3 m landing area/MARV out of which 50% is the suction surface. Finally, a brief cost analysis has been carried out ^[T1], considering the unit cost of three NASA autonomous cars (75,000/unit) and approximately 50,000\$ for retrofits, resulting in a total MARV cost of 275,000\$.

Constituents	Mass Estimation (kg)	Estimated Mass Limit (kg)	Weight Fraction
Airframe	199.576	289.4	0.52
Solar Panels	62.1	62.118	0.11
Systems & Electronics	20	20	0.04
Powerplant	33	60.1	0.11
Fuel Cells	265	107.76	0.19
Payload	20	20	0.04
Total	599.7	559.4	1

Table 4: Initial weight and weight fraction estimates.

4. Vehicle Weight & CG Determination

4.1 Initial Weight Estimate

The initial mass and weight fraction estimates that form the result of the first 3 iterations weight analysis are summarised by Table 4. These were obtained through a combination of empirical formulas, historical data and the use of constraint diagrams. These techniques were especially useful for the first iterations of the A/C, however, by the second iteration a more accurate model has been developed, which proved consistent with the estimates. From this point forward the ‘relay had been passed’, the accurate model becoming a design-driver.

4.2 Accurate Mass & CG Model

The aim of this model was to offer a centralised system that accounts for the mass and CG location of each individual component of the A/C. It enabled quick modification of all parameters and output of an accurate global CG location and total A/C mass. From the beginning the model was designed with the ease of increasing mass discretisation methods in mind. As can be seen from the table below, main parameters such as the wing, were divided into individual subcomponents. Due to the complex nature of the interaction with other teams (summarised in a map included in Annex D), input ease and output clarity were essential. It is important to note the interdependency of the mass discretisation and CG location accuracy the position of a component, the structural mass associated with the new position will change the total system mass, in turn changing the CG again. This is the reason why the model has to consider both A/C mass and CG.

4.3 Output of the script

Figure 8: Weight estimate evolution for the Everplane concept across all iterations featuring the use of the accurate model from week 5 onwards.

Component	Subcomponent	kg	Xg (m)				Yg (m)	Zg (m)
Engines	Electromotor	0.70	11.655	15.655	-15.655	-11.655	-1.000	0.000
	Propeler & Cone	4.30	11.655	15.655	-15.655	-11.655	-1.000	0.000
	Engine Shaft	0.95	11.655	15.655	-15.655	-11.655	-0.500	0.000
	Total	5.95	0				-0.920	0
Power	Solar Panels	51.29	0				0.882	0.086
	Pods Structure	18.48	27.310		-27.310		-0.032	0.000
	Pods Content	204.00	27.310		-27.310		-0.032	0.000
	Total	273.77	0				38.244	0.016
Systems	Avionics	2.84	0				0.200	-0.109
	Cables	11.44	0				0.850	0.109
	Actuators	2.52	0				0.200	-0.109
	Total	16.80	0				0.642	0.039
Payload	Antennae + Transmitters	1.17	20.43, 14.43, 8.43, 14.43, 14.43, 0				2.878, 2.978, 2.878, 1.214, 13.91935	0.000
	Spare Space for Module	18.83	0				0.200	-0.109
	Total	20.00	0.704				0.468	-0.103
Main Wing	Skin	9.40	0				0.882	0.026
	Ribs	101.11	0				0.882	0.026
	Main Spar	74.00	0				0.850	0.109
	Joints	1.38	0				0.882	0.026
	Control Sufaces	1.02	0				2.428	0.026
	Total	186.90	0				0.878	0.059
Tail	Horizontal & Vertical	8.61	0				14.416	4.826
	Fuselage	12.61	0				6.750	0.500
	Solar Panel (not used)	0.00	0				14.961	0.086
	Total	21.23	0				9.860	2.255
Global		539.65	0				0.760	0.120
34.962 % of MAC								

Table 5: Complete analysis of the A/C mass and global CG location (to be used in conjunction with the axis definition in Annex D.

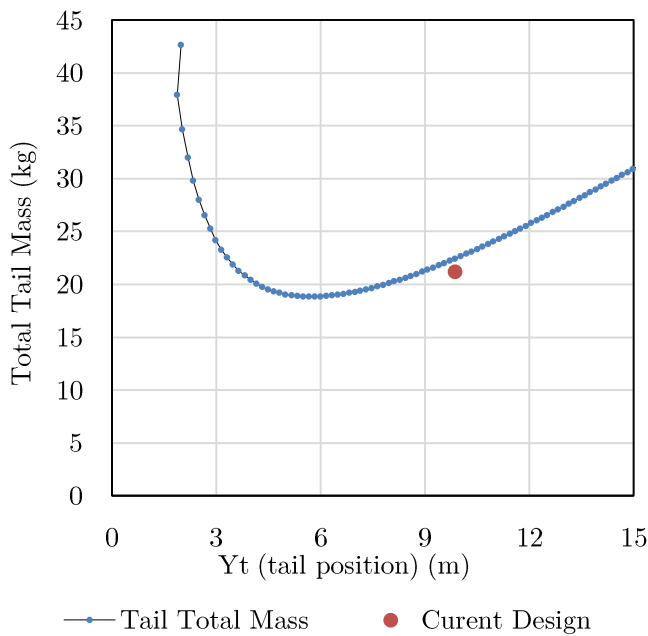


Figure 10: Tail CG position vs. Tail Mass

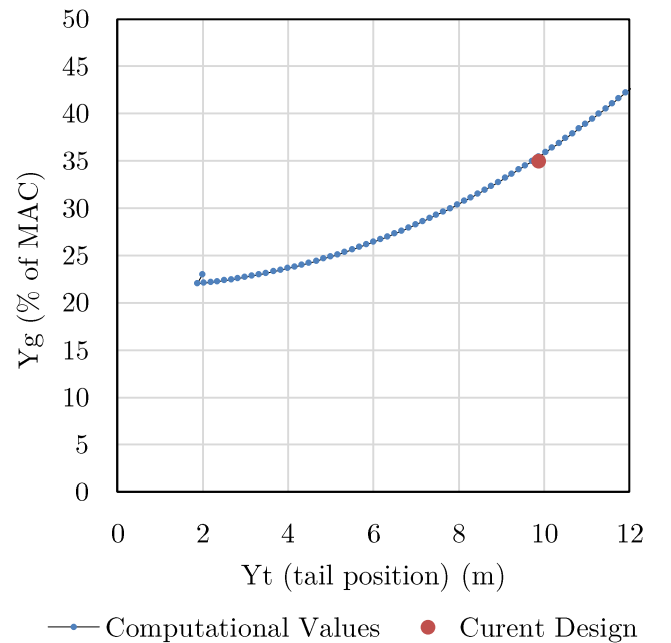


Figure 9: Tail vs. Global C.G. position

A further discretisation of the wings, pods, fuselage, vertical and horizontal tail components and payload is included in Annex D. Figures 9 and 10 were used throughout the design in collaboration with the Flight Dynamics to establish an optimum tail design. Because the tail position affected tail mass and in turn global CG location, an iterative process was necessary. The optimum design is also highlighted on the figures and represents the intersection of the neutral point variation curve with respect to tail position and global CG variation with respect to tail position.

4.4 Limitations of the code

It is important to keep in mind the assumptions made throughout the model as they have a significant impact upon the results. First of all, the script is limited by the level of discretisation used, an even more accurate model can be created by discretising the parts further. Other significant assumptions are:

- Assumed perfect symmetry with respect to the x axis ($x_{CG}=0$).
- The geometry of some parts was simplified in order to determine CG.
- The fuselage is assumed to have a CG at mid-length, so are engine shafts.
- The propeller, engine and cone are assumed to have the same CG.
- Mass of engines is estimated from regression data.
- Ribs mass is spread equally along the skin of the main wing.
- For the vertical and horizontal tail there is a similar situation with spar also smeared.

For preliminary design these approximations are acceptable and greatly simplify the process, allowing for quick multiple iterations. However it is important to note that for the detailed design phase these issues need to be addressed and an appropriate safety margin should be implemented subsequently. Currently a safety margin of 10% was taken for the mass. The initial mass estimate being 559.4kg and the final mass being 539.65kg (lower by 4.36%).

5. Computer Aided Design

5.1 Use throughout design

Creo Parametric 3.0 was used to create aircraft parts and assemblies, Simulate 3.0 was used for preliminary structural analysis, Abaqus CAE for detailed analysis ^[T4] and Star CCM+ for CFD aerodynamic optimisation [T3]. CAD was used throughout design stages, using the procedures recommended by Raymer [6]. Concept sketches ([Annex B](#)) were created for every vehicle configuration analysed. The use of Creo Parametric to determine a more accurate value for individual parts mass and global CG was assessed establishing that for detailed design this would be a viable option, as long as accurate A/C sizing and material selection is provided by the other subgroups.

5.2 Technical Drawings

For the latest design iteration, drawings of the general assembly and main constituent parts have been created in accordance with ISO standards [7]. In the interest of space these have been included in Annex H.

5.3 Renders and visual presentation

With the use of the specialized Autodesk 3DsMax 2016 software package and the V-Ray rendering engine, the final versions of the CAD parts were processed and to obtain photorealistic images (Annex G). The processed involved the import of files from Creo 3 Parametric, material definition, assignments to parts, rendering environment setup (lighting and camera parameters) and finally static images and animations generation. By this means the cover and poster images (which involved over 12 mil. tessellation elements) were created.

6. Conclusion

6.1 Accomplishments

Having determined the optimum vehicle configuration for the mission requirements and having established a total system weight and global CG location it is possible to conclude that this Everplane configuration is capable of achieving mission requirements. Computer Aided design has been successfully used through the project enabling a faster analysis aircraft configurations and finally a realistic way of representing the concept.

6.2 Future Improvements

If sustainability was not one of the priorities, the aircraft could be made significantly lighter. Preliminary estimations on the structural mass improvements ^[T4] suggest a 30kg reduction. Also, it is important to note that the A/C is capable of carrying an additional 18.83 kg of modular payload for extended mission objectives such as aerial imaging in addition to the mass required to achieve its primary mission. This can be reduced as well, if not required. Refining the CAD model to include materials in order to accurately establish a CG position and in turn the exact position of each component relative to others could be implemented. Taking into consideration aero elastic effects given by the inherent flexibility of the high AR wing, initial research was carried out in this matter, specifically for HALE A/C [18] [19], in order to establish its impact on vehicle definition, however for preliminary design there was not sufficient time to analyse this issue due to its complexity. In the end, the feasibility of the Everplane Vehicle definition has been confirmed and it can be further sustained by additional iterations and detailed design. Therefore, if constructed, the aircraft will set new for autonomous high altitude long endurance aircraft in a sustainable manner, providing an aerosattelite platform.

Internal References

Notations T1-T5 in the report refer to direct contribution of the following teams:

T1 – Mission Definition Team

T2 – Another member of the Vehicle Definition Team

T3 – Aerodynamics Team

T4 – Structural Analysis Team

T5 – Payloads and Systems Team

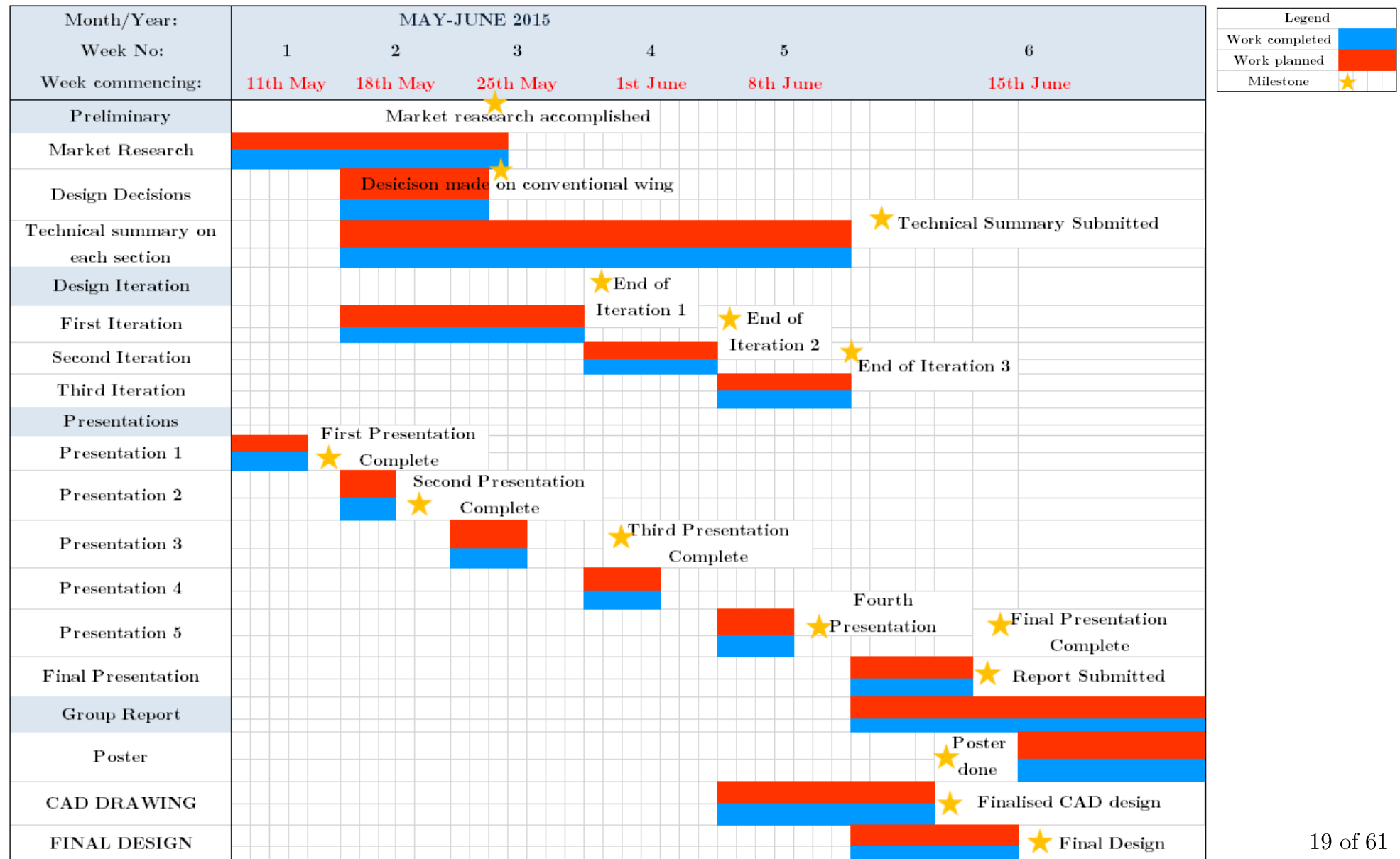
T6 – Propulsion Systems Team

T7 – Flight Dynamics & Controls Team

External References

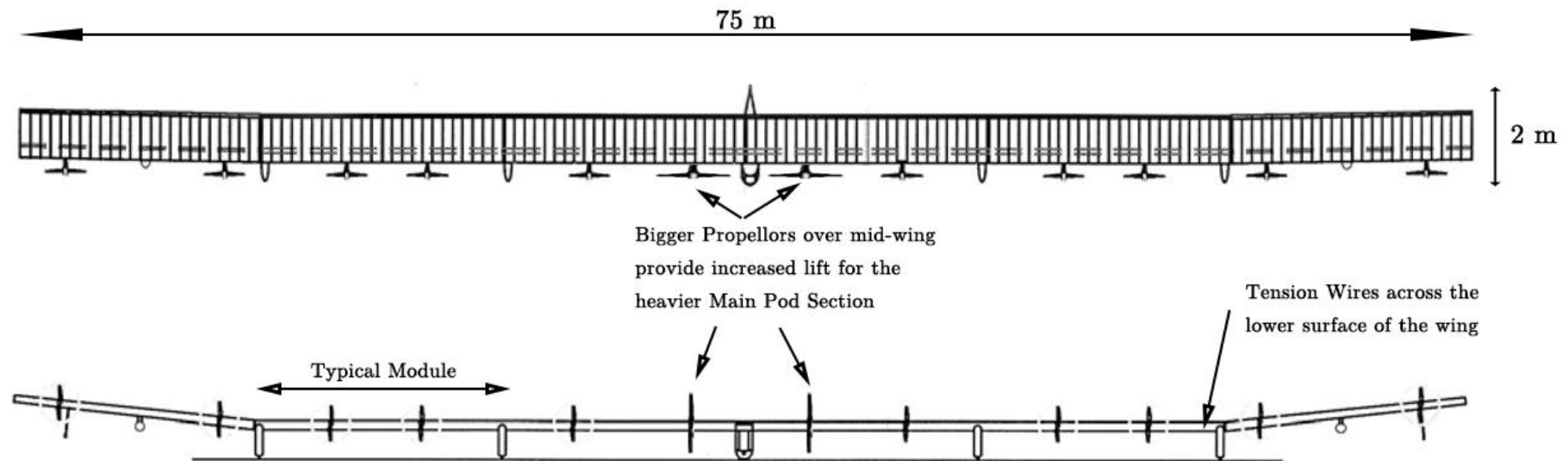
[1] T. Megson, "Aircraft Structures for Engineering Students," Elsevier Science, 2012.

Annex A: Project Timeline - GANT Chart ^[T1]



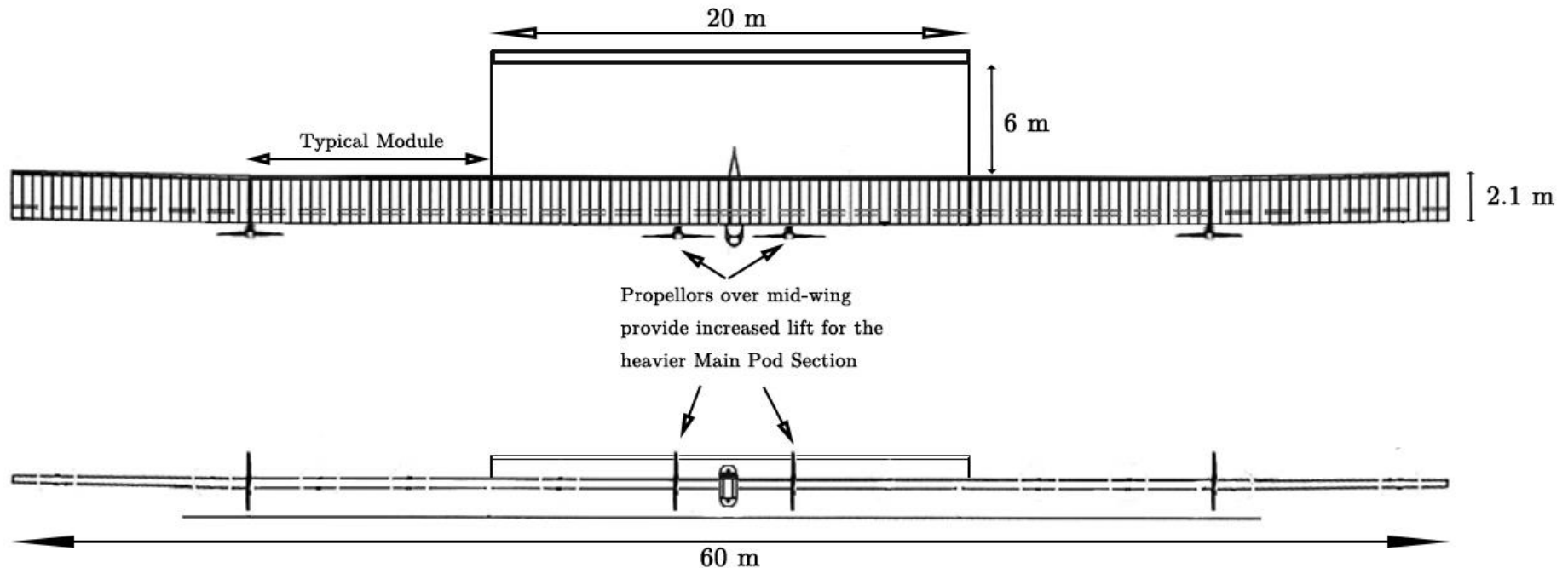
Annex B: Sketch Iterations Log

Everplane V1.01



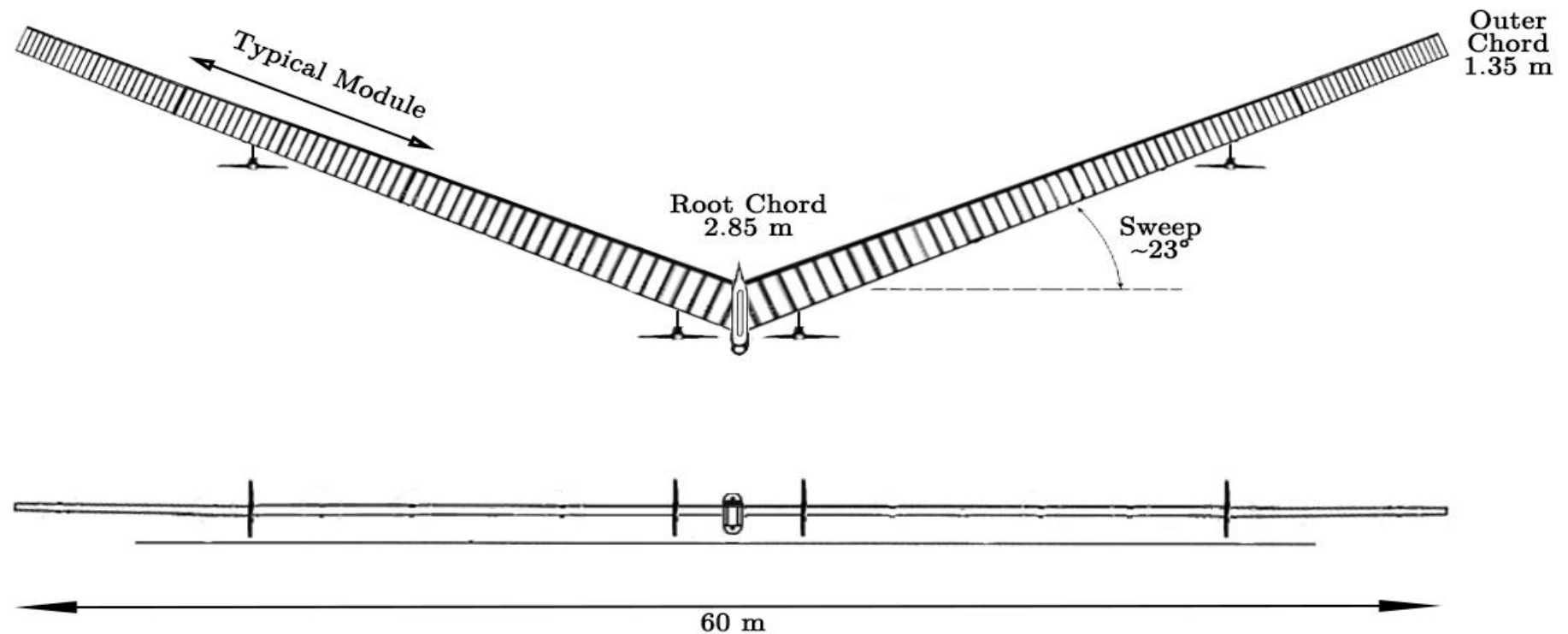
- The very first iteration, presents the baseline configuration we established after 3 days work.
- Distributed propulsion all along the wing. Idea to have propellers close to the wing.
- One main pod containing fuel cells / no pod and distributed batteries in the wing.
- Landing gear incorporated in the wing.
- Exploring the idea of differential lift behind propellers close to the main pod.

Everplane V1.02- 1 main pod version



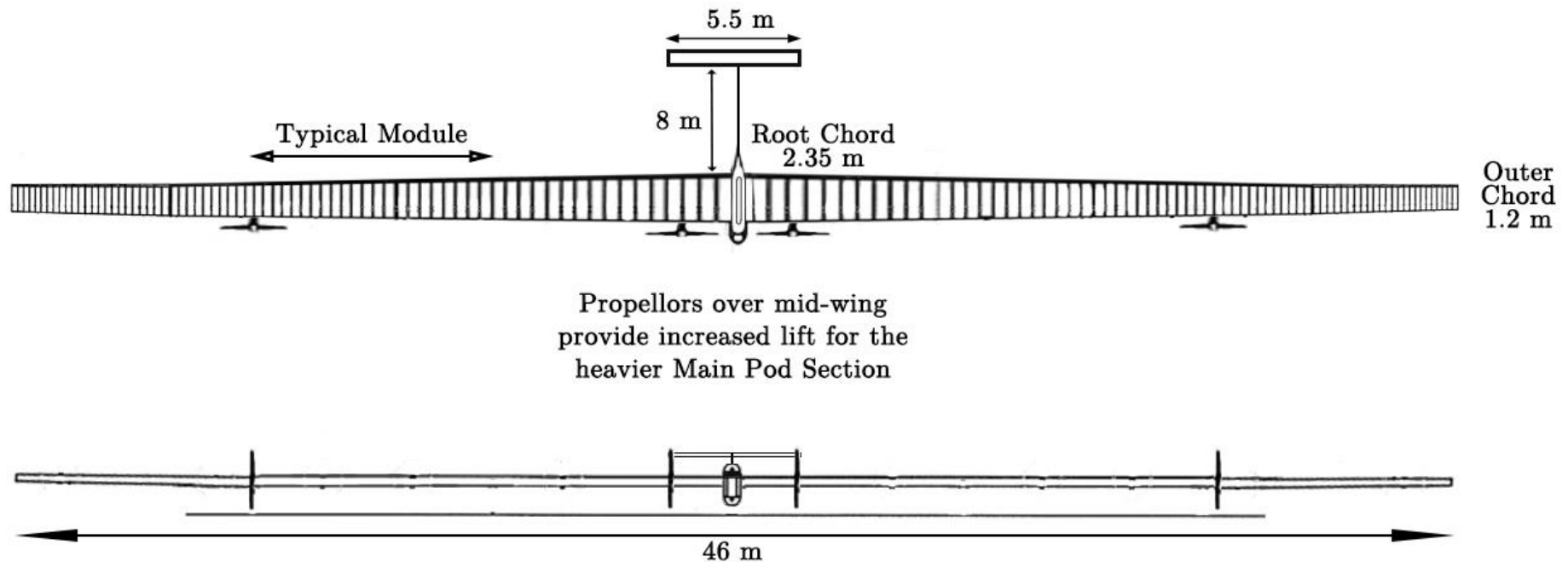
- Initial tail added, offering only pitch authority.
- Dimensions of control surfaces are only provisional.
- Propulsion system reduced to 4 engines with 2 m diameter propellers.
- Differential thrust provides yaw control.
- Chord increased to 2.1m.
- Span decreased by 15m to 60m, after evaluating the first constraint diagrams.
- Initial tail added, offering only pitch authority.

Everplane V1.03 - Swept & Tapered



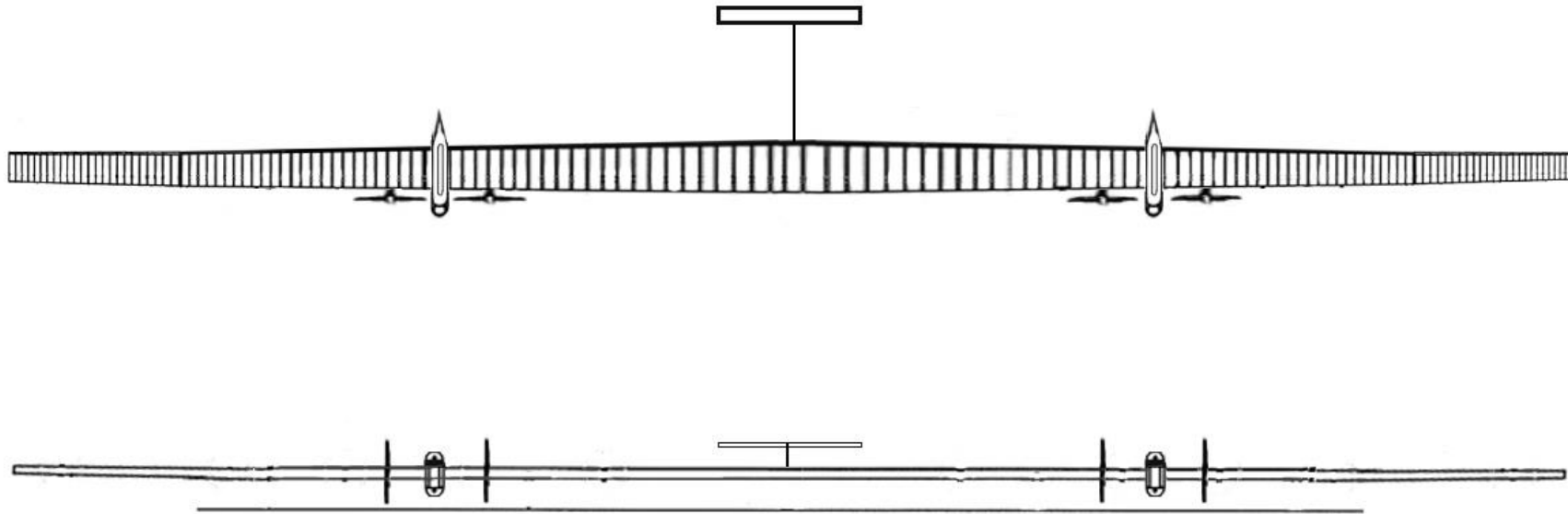
- The focus of the second week was the comparison of a sub-iterations 1.03 (swept) and 1.02 (conventional).
- Sweep angle is only provisional, established by the Vehicle Definition team based on interpolated historical subsonic A/C data.
- The weight of this configuration is under evaluation.
- Assuming the same main wing, respectively taper and span.

Everplane V2.01



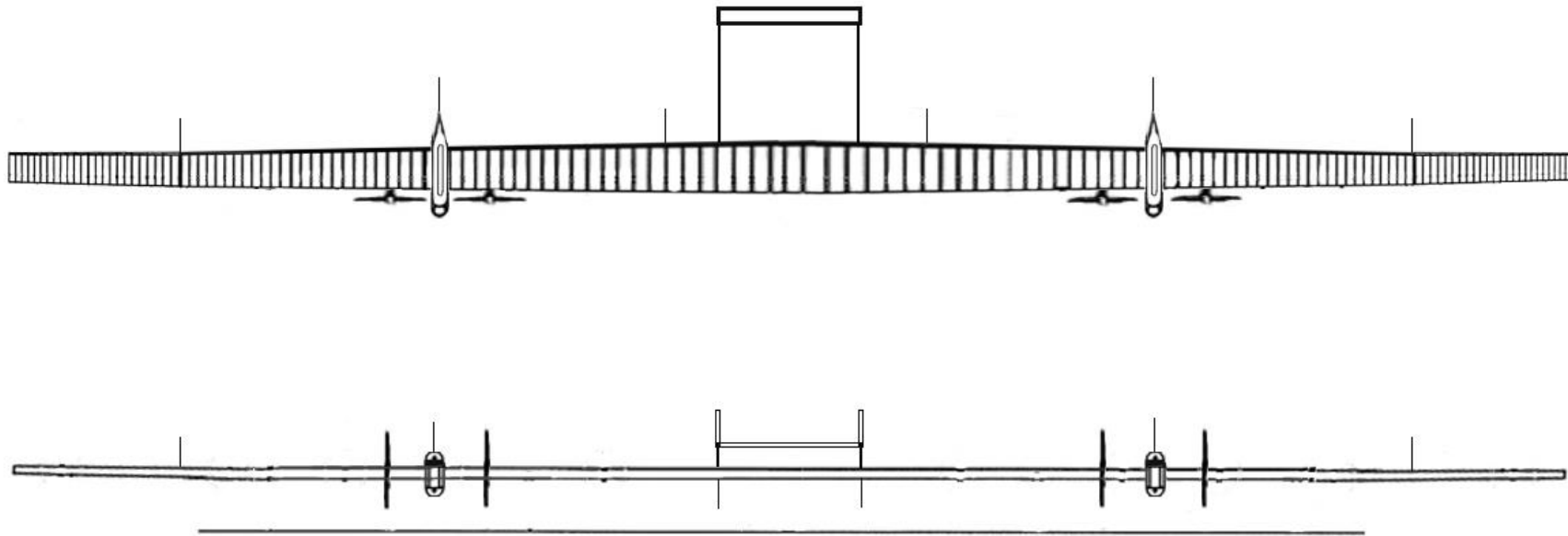
- Third week iteration.
- Single boom tail was found to be more structurally efficient than a twin boom design.
- Dimensions of control surfaces are only provisional.
- Horizontal tail elevation anticipated to avoid main-wing wake.
- A single main pod will contain hydrogen fuel cells. Battery option dismissed.
- Taper added from 2.35m to 1.2m.
- Span decreased by 14m to 46m, after evaluating the second constraint diagrams.

Everplane V2.02



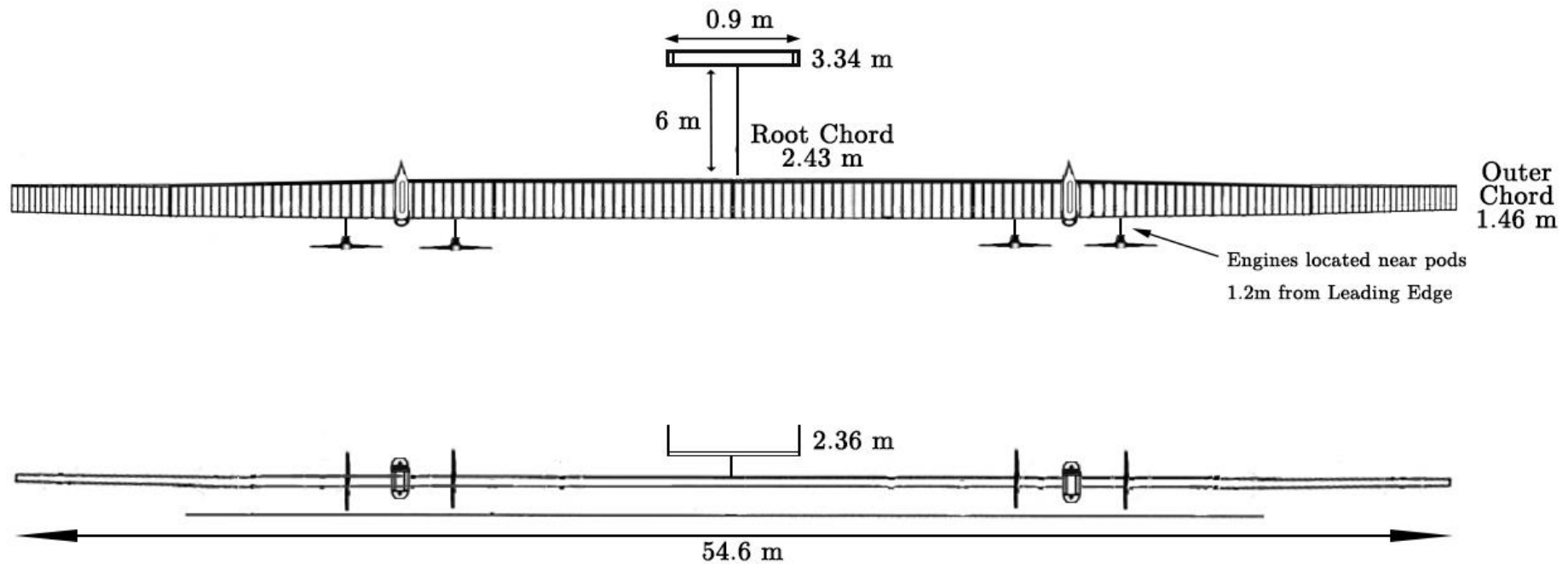
- Sub-iteration considering another distribution of the fuel cell system.
- Landing gear entirely removed.
- Wing not capable of sustaining self-weight.
- Main wing and tail unchanged.

Everplane V2.03



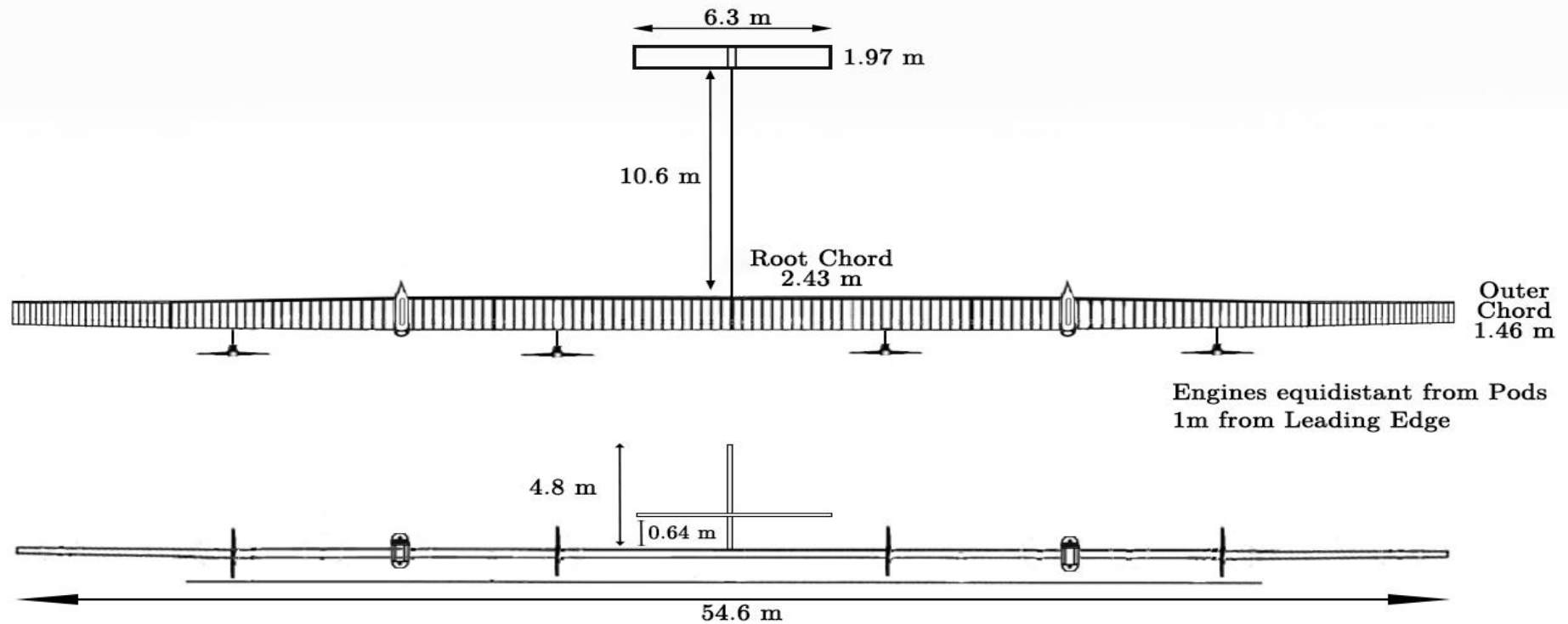
- Third week sub-iteration.
- A more refined weight model has been developed and has become the design-driver.
- Systems and Payloads have been included in the mass model (antennae's preliminary locations shown)
- Wing remains unchanged, but mass has been updated, resulting in an outdated wing design.
- Twin boom tail design to be analysed from a stability point of view to determine if it overrules the structural disadvantage.

Everplane V3.01



- Configuration at the start of week four.
- Tail design is still preliminary, but all previous alternatives have been considered.
- Final wingspan established as 54.6m
- Final taper determined.
- Two pods at $\pm \frac{s}{2}$ found to be the optimum from a structural point of view.
- Engine Chordwise location established, Spanwise location under evaluation.

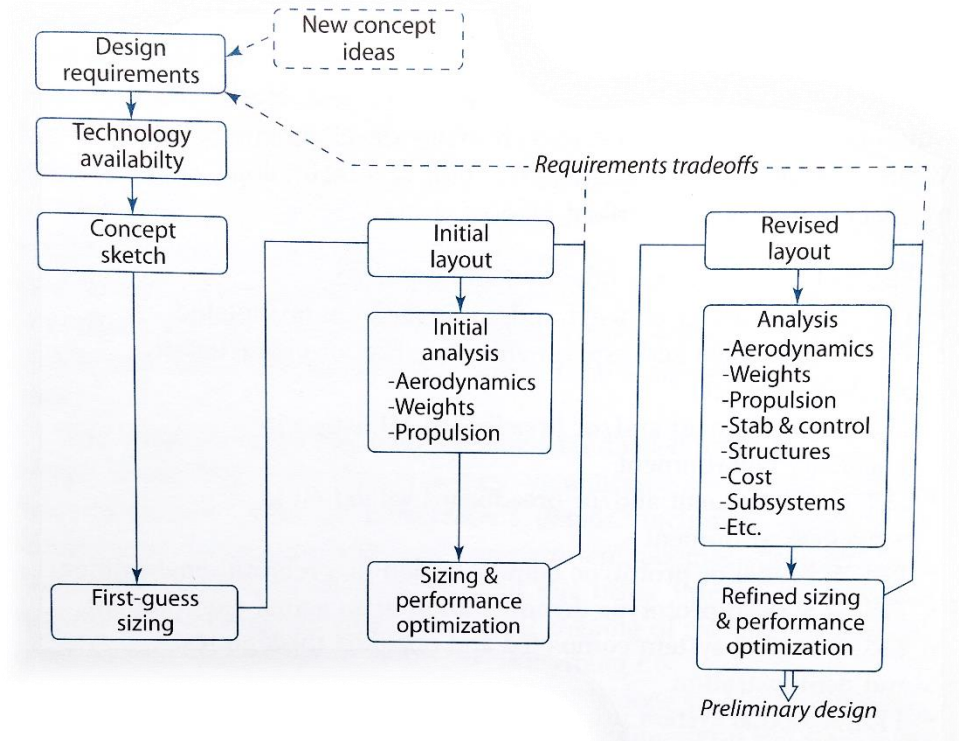
Everplane V3.02



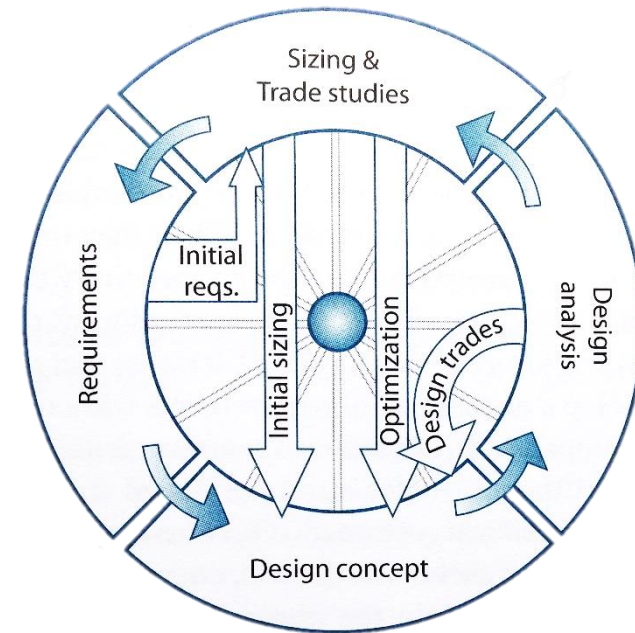
- Week four final configuration.
- All parameters except tail dimensions have been established.
- Engine and main pod optimum location has been determined.
- The mass of the main pods increased by over 80% in order to be able to climb at sea level, this was due to a miscalculation of the Propulsion Team. Vehicle definition and Structures team to assure a sufficient weight reduction to be within parameters.
- Due to sufficient safety margin in the mass calculation, the wing design was still valid.

Annex C: Aircraft Conceptual Design Strategy

The following diagrams were followed throughout the design cycle, according to the methodology detailed by Raymer [10].



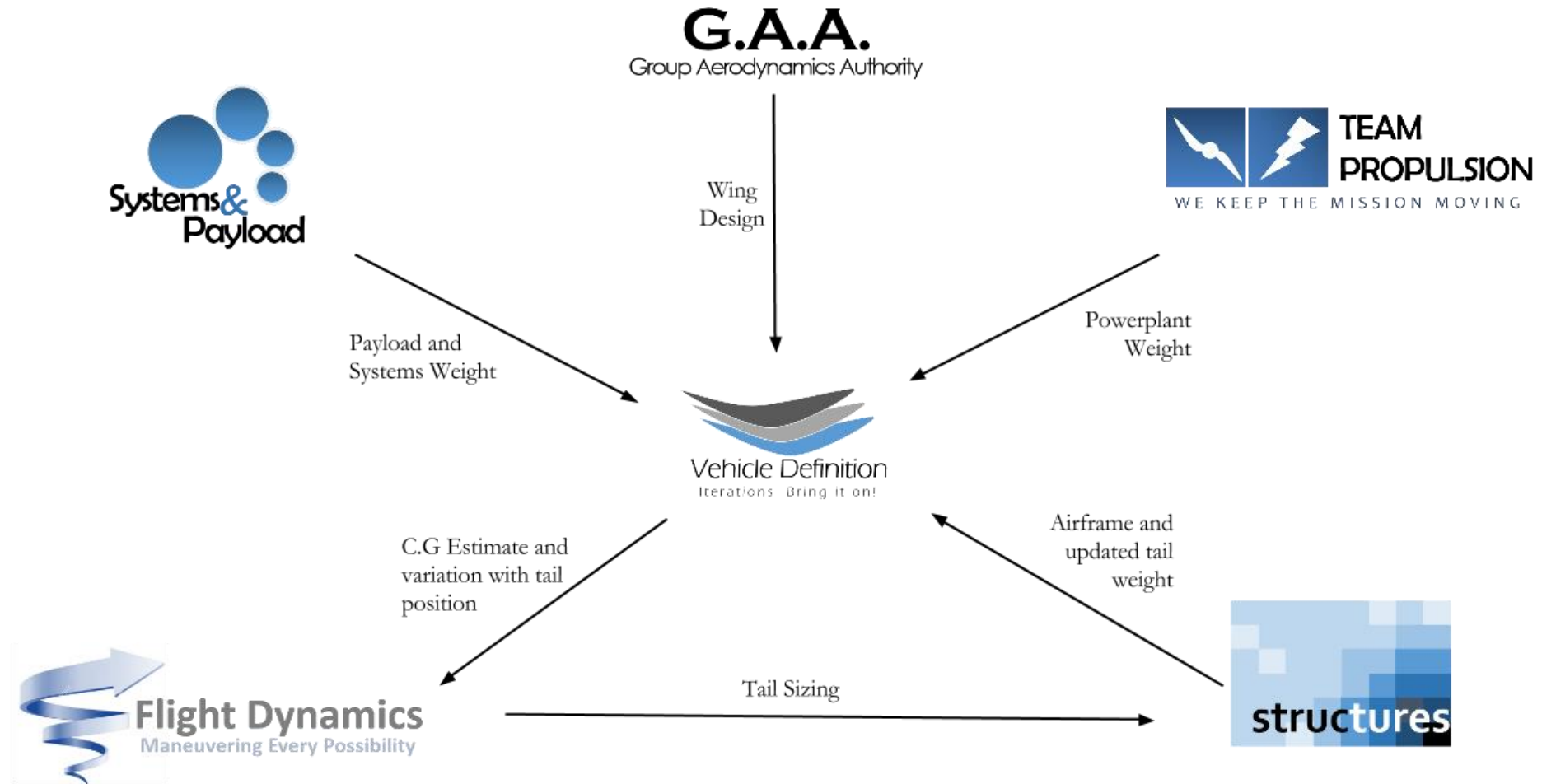
Aircraft conceptual design stages and their relationship [10].



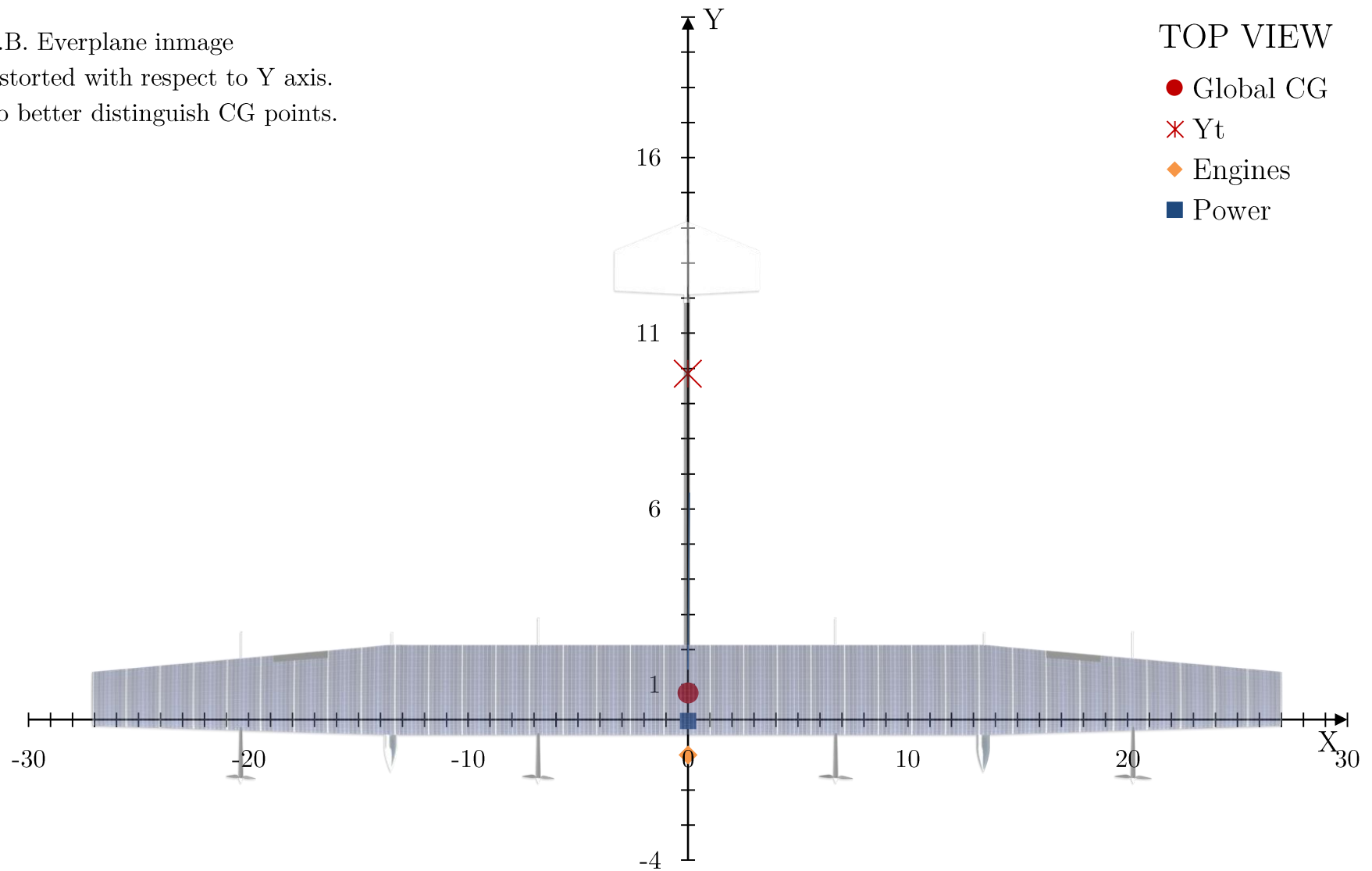
Design wheel for one iteration [10].

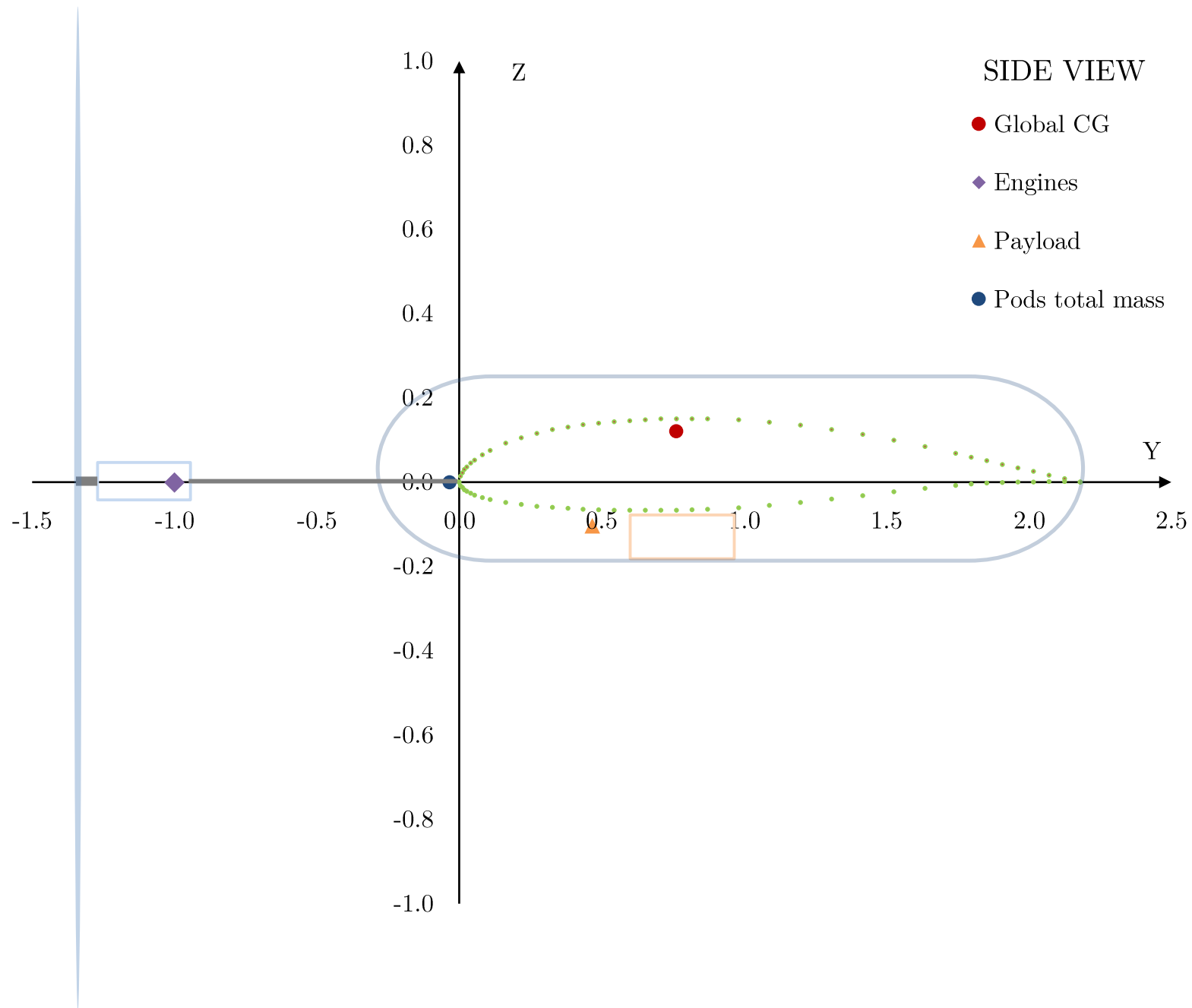
Annex D: Interaction Map

The following graphic shows the centre of gravity & system mass inputs and outputs required from/by other teams.



N.B. Everplane image
distorted with respect to Y axis.
To better distinguish CG points.





Further mass discretisation ^[T4]

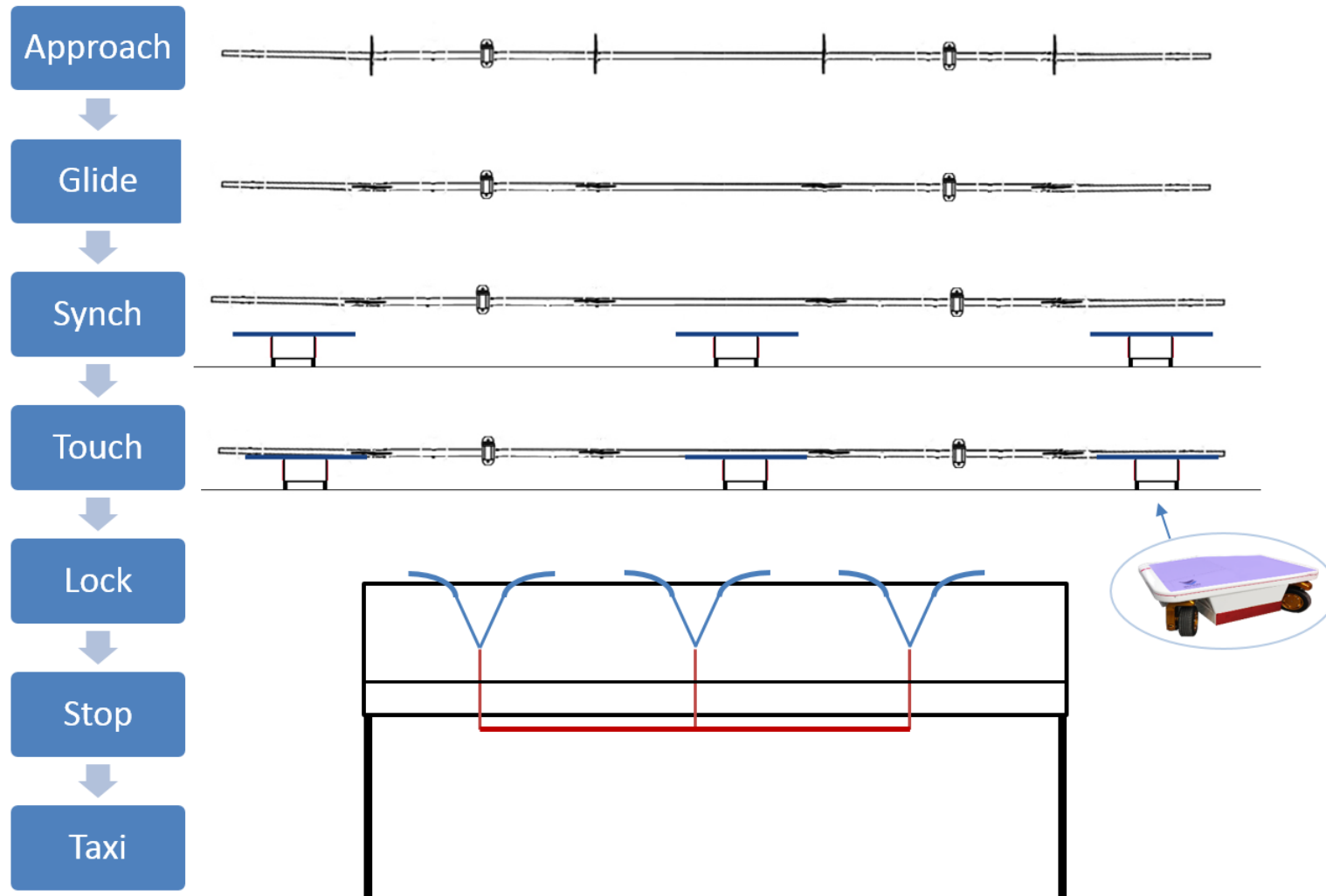
Fuselage	
Component	Total Mass (kg)
Booms	11.35
Ties	0.38
Frames	0.24
Skin	0.644
	12.614

Pods	
Component	Mass/ pod (kg)
Skin	1.35
Heavy Frame	1.102
Light Frame	0.009
Total Pod	4.94
Bulkhead	1.581
Trusses	0.267
	9.249

Tail		
Component	Vertical Tail	Horizontal Tail
	Mass	
Spar	0.318771	0.935518
Ribs	1.138	2.36
Skin	1.3072	1.40929
Control Surfaces	0.588	0.55679
	3.351971	5.261598

Payload			
Avionics	Weight (per unit)	Units	TOTAL Weight
Motherboard Computers	0.018	18	0.324
Transponder	0.1	5	0.5
GPS chip	0.1	5	0.5
EPMU	0.2	3	0.6
IMU chip	0.1	5	0.5
Thermocouples	0.012	35	0.42
Wiring	Weight (per metre)	Metres	
14 SWG Copper Wire	0.02	572	11.44
Total Weight		14.28	kg

Annex E: MARV Landing Procedure



Annex F: Balloon Sizing Script

```
function [Rho,P,T] = Atmosphere(Alt)
% This function takes input 'altitude' [km] and interpolates using
% tables of atmospheric data to find parameters

% Find value from tables
load 'atmospheric.mat'
j=1;
while Alt > density(j,1)
    j=j+1;
end

Rho = ((Alt-density(j,1))/(density(j+1,1)-density(j,1)))*(density(j+1,2)-
density(j,2)) + density(j,2);
P = ((Alt-density(j,1))/(density(j+1,1)-density(j,1)))*(density(j+1,3)-
density(j,3)) + density(j,3);
T = ((Alt-density(j,1))/(density(j+1,1)-density(j,1)))*(density(j+1,4)-
density(j,4)) + density(j,4);
```

```
%Name: Balloon size & Cost
%Goal: Determine a first approximation of the balloon size and associated cost
to launch at a given altitude. Plot variation of cost and balloon size w.r.t
desired altitude.
%Created by: SERGIU PETRE ILIEV. Date: 10.06.2015
%Imperial College London | Aeronautical Engineering | GDP | Coursework
close all; %clean workspace
clc;
clear all;

%% Assumptions
% consider just the limiting case, when the balloon reaches desired altitude and
has to sustain all mass; neglect ascent time, drag on envelope and A/C

%% Initialize parameters
fprintf('Inputs \n');
h=[20,0:0.1:30]; %first value is the input height,
following values are from 0 to 30km in steps of 100m
% h(1,1)=input('Desired altitude to be reached by the balloon (km): ');
m=600;
% m=input('Aircraft total mass (kg): ');
k=2.77; %$/kg of liquid H2 (cost)

%% Researched Data, source: www.redbullstratos.com/technology/high-altitude-
balloon/
t=2.03200*10^-5; %m thickness of balloon envelope skin
(based on Red Bull Stratos)
    %Polyester-fiber reinforced load tapes are incorporated for weight bearing
    %Balloon mass = of skin + estimated mass of reinforcements wires, parachute,
balloon avionics, release valve nylon "destruct line" will release the helium so
that the balloon returns to Earth. Then, the team will gather the envelope into
a large truck, to be inspected at the base and reused.
q_material=889.6; %kg/m3 smeared density of balloon
V_stratos=849505.398; %m3
R_stratos=(849505.398*3/4)^(1/3); %m
m_balloon_stratos=q_material*t*4*pi*R_stratos^2; %kg mass of the entire balloon
%knowing the actual mass of the Red Bull Stratos as 1680kg, a loop was done to
find the smeared density above that would give m_balloon_stratos=1680kg
```

```

%% Initialize
% find the AIR atmospheric parameters for all heights using a function that
interpolates experimental data NB: better results can be obtained is the
interpolation is not done between 1km height values
for i=2:302 %from 2 (to skip the input value) to 302 (size of h)
[Rho(i),P(i),T(i)] = Atmosphere(h(i)); % air density, pressure and temperature
end

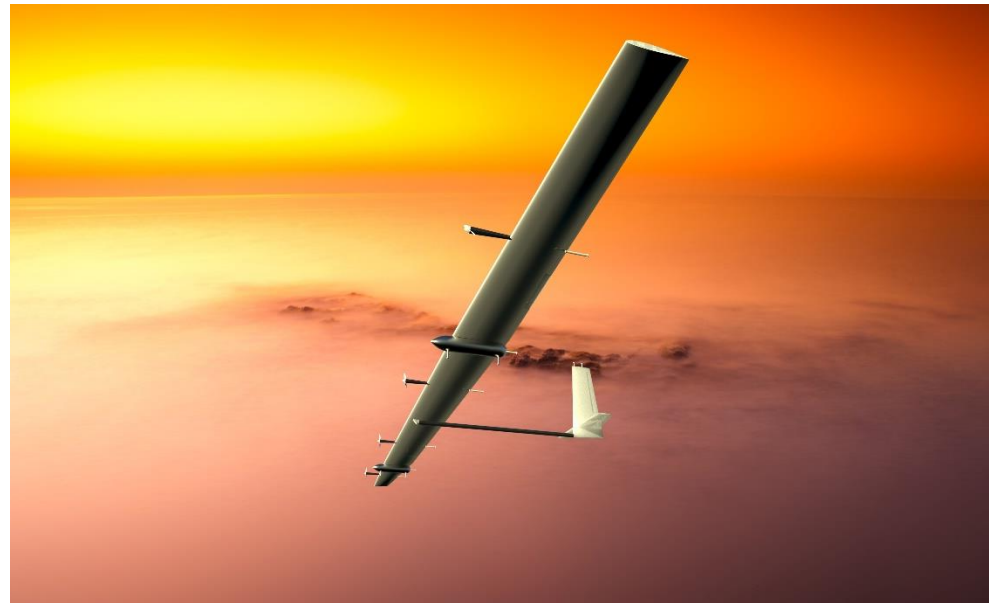
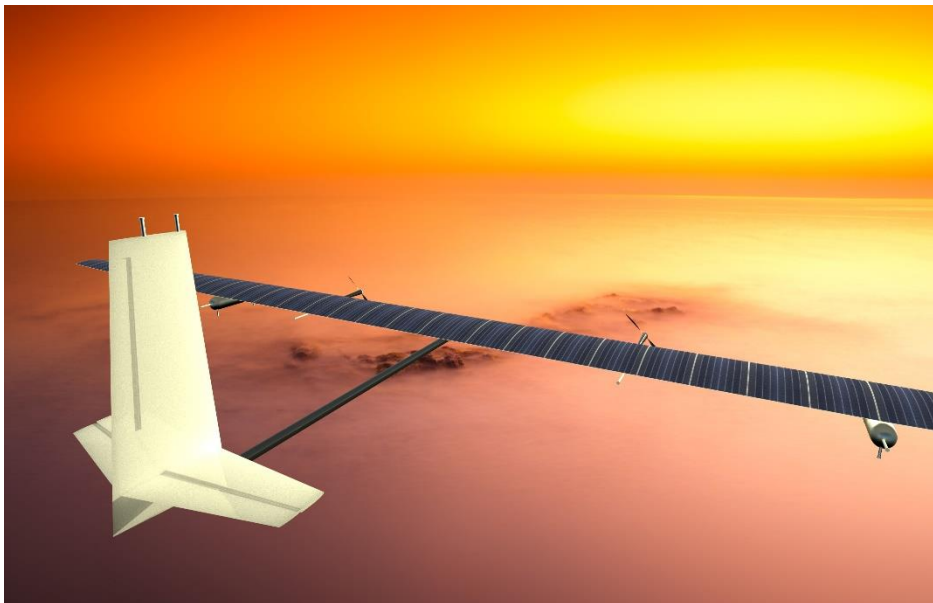
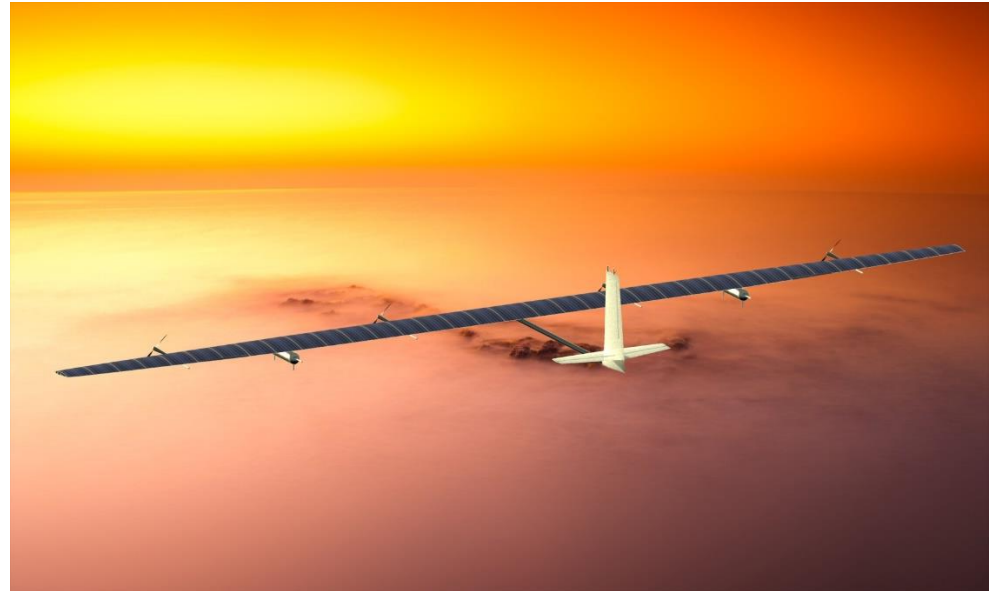
%% Loop to stabilize mass of the system for all possible altitudes
%acceptable error of 100 grams
% find the H2 atmospheric parameters as a function of AIR atmospheric parameters
(assume H2 and AIR behave the same (same Cp)
    % using H2 because it has a lower density, is cheaper, but has the
disadvantage of escaping the envelope over a long time
m_total(1)=m + 15+50+54.57+2+50; %15kg wires and quick release system, 50kg
vaporiser, 54.57±0.1kg of H2, 10kg recovery parachute and 2kg control unit
m_balloon(1)=0; %assume massless baloon initially
eps(1)=20;
for j=2:302
    c=2;
    while eps(c-1)>0.01 && c<100 %within 10 g not more than 100 iterations
        rho_h2(j)=0.0899*( (P(j)/P(2))*(T(2)/T(j)) );%kg/m3 hydrogen density at sea
level * from perfect gas law ratios = hydrogen density at various heights;
assuming it adapts perfectly to temperature and pressure of the exterior
        R(j)=((3*m_total(c-1) )/( 4*pi*(Rho(j)-rho_h2(j)) ))^(1/3); %m balloon
radius at altitude
        V(j)=(4*pi/3)*R(j)^3; %m3 volume of balloon at altitude
        m_balloon(c)=1.5*q_material*t*4*pi*R(j)^2;%50% safety margin
        m_total(c)=m_total(1)+m_balloon(c);
        eps(c)=m_total(c)-m_total(c-1);
        m_balloon_save(j)=m_balloon(c); %remember balloon mass
        c=c+1;
        Cost(j)=k*V(j)*rho_h2(j)*50; %$ H2 cost associated with 50 Everplanes
    end
    m_total(j)=m_total(c-1);
end

%% Plot
figure
subplot(1,2,1)
plot(m_balloon_save(2:end), h(2:end), 'b') %plot the experimental response
xlabel('Baloon Mass (kg)'); %label the vertical axis
ylabel('Desired insertion (i.e. neutral buoyancy) altitude (km)'); %label the
horizontal axis
% plot(R(2:end), h(2:end), 'b') %plot the experimental response
% xlabel('Balloon Radius (m)'); %label the vertical axis
% ylabel('Desired insertion (i.e. neutral buoyancy) altitude (km)'); %label the
horizontal axis

subplot(1,2,2)
plot(Cost(2:end), h(2:end), '--r') %plot the reference signal
xlabel('Fleet launch cost ($)'); %label the vertical axis
ylabel('Desired insertion (i.e. neutral buoyancy) altitude (km)'); %label the
horizontal axis

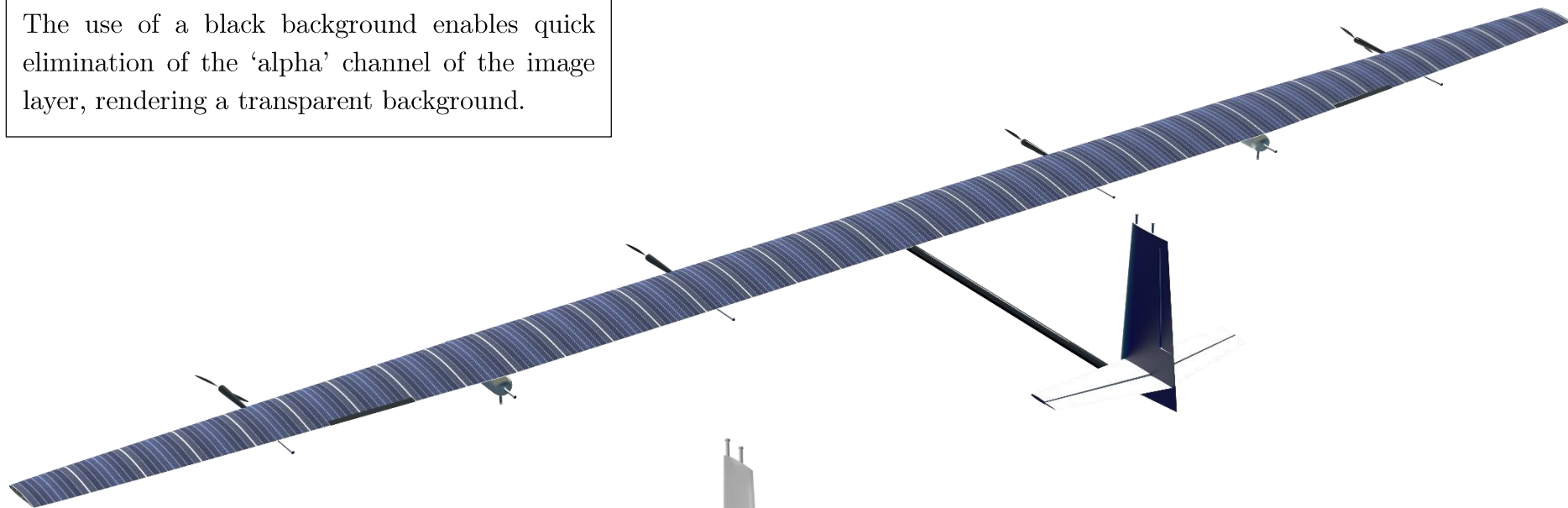
```


Annex G: Photorealistic renders

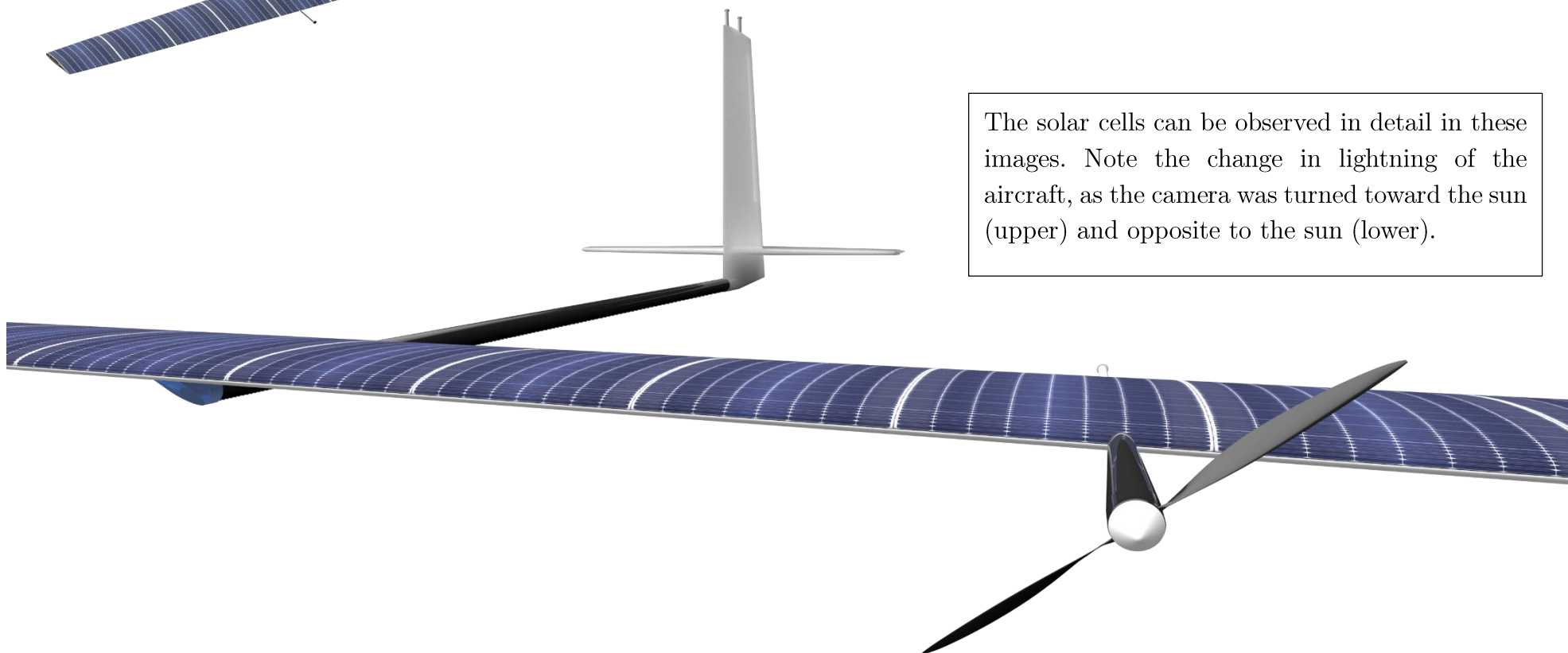


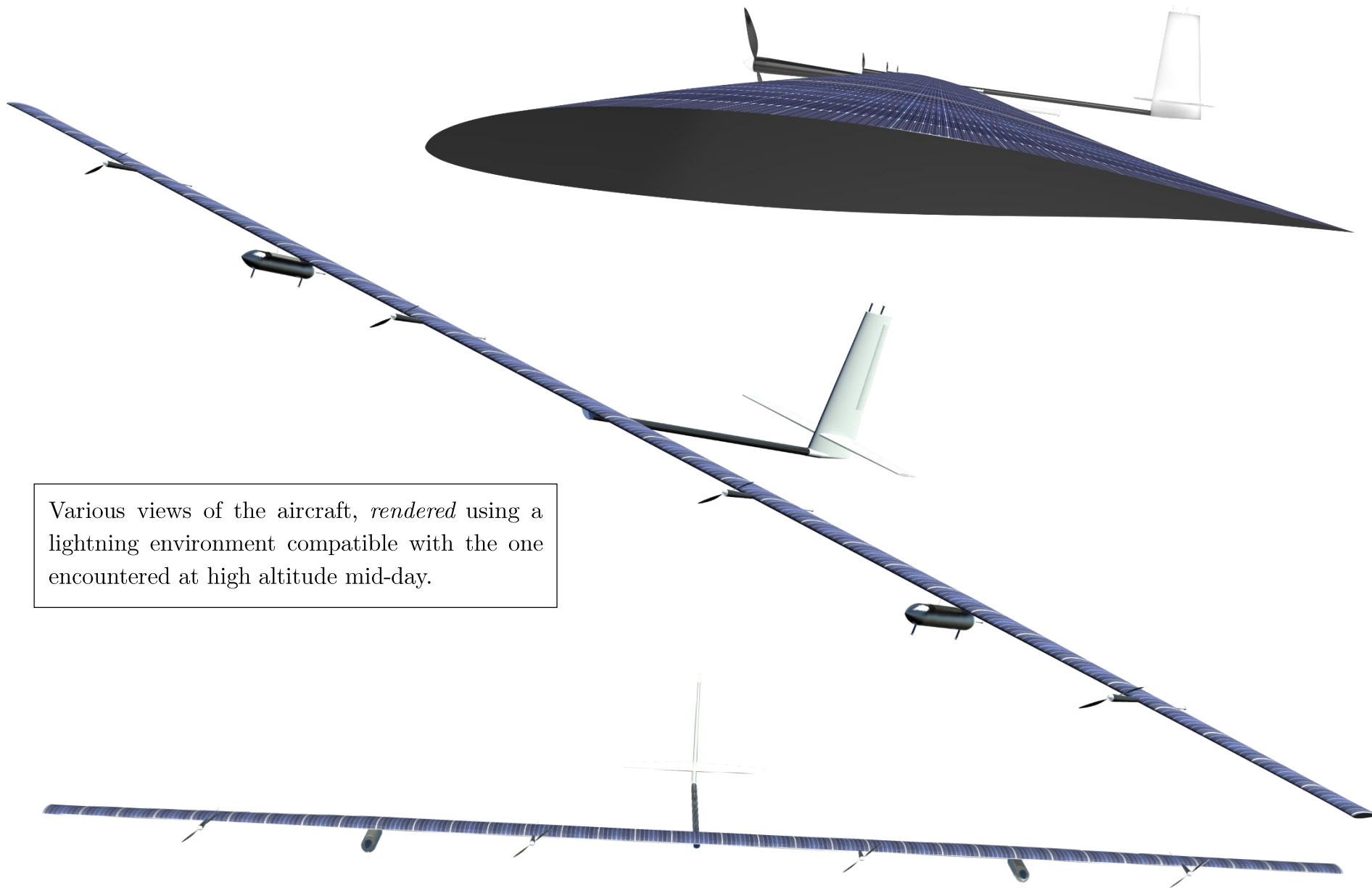
The sun position and lighting parameters (color, intensity, and angle) can be reproduced in the rendering environment, specific to the time of day.

The use of a black background enables quick elimination of the 'alpha' channel of the image layer, rendering a transparent background.

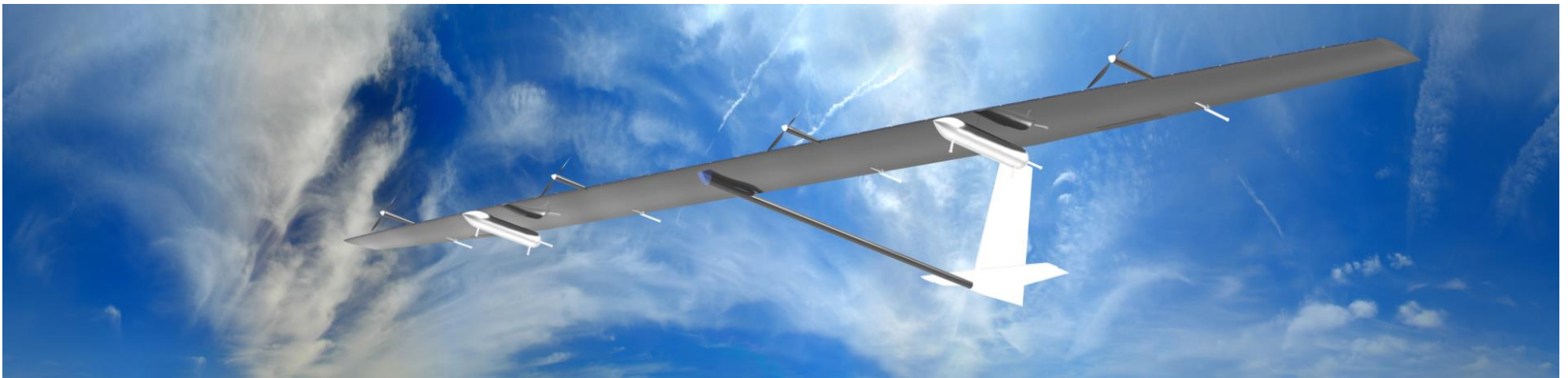
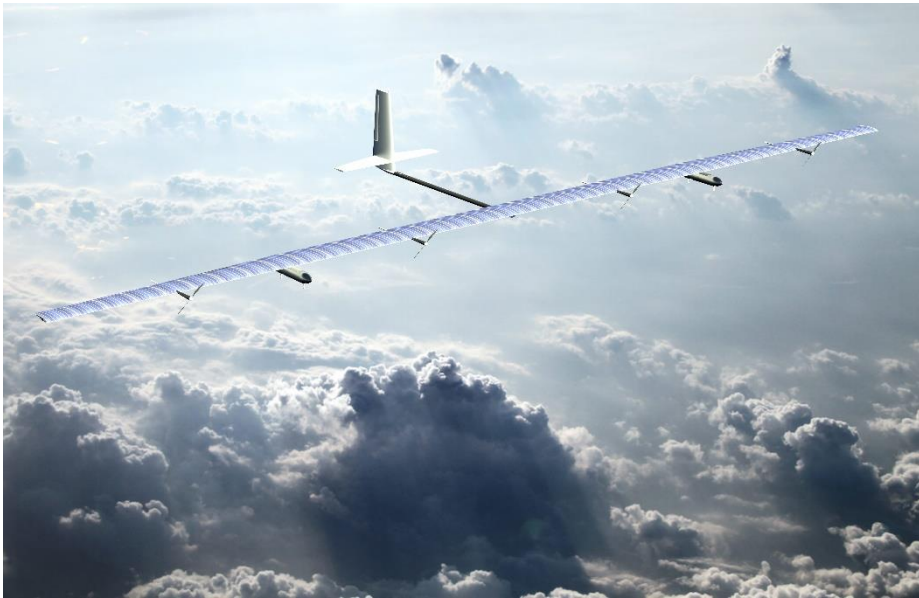


The solar cells can be observed in detail in these images. Note the change in lighting of the aircraft, as the camera was turned toward the sun (upper) and opposite to the sun (lower).





Various views of the aircraft, *rendered* using a lightning environment compatible with the one encountered at high altitude mid-day.



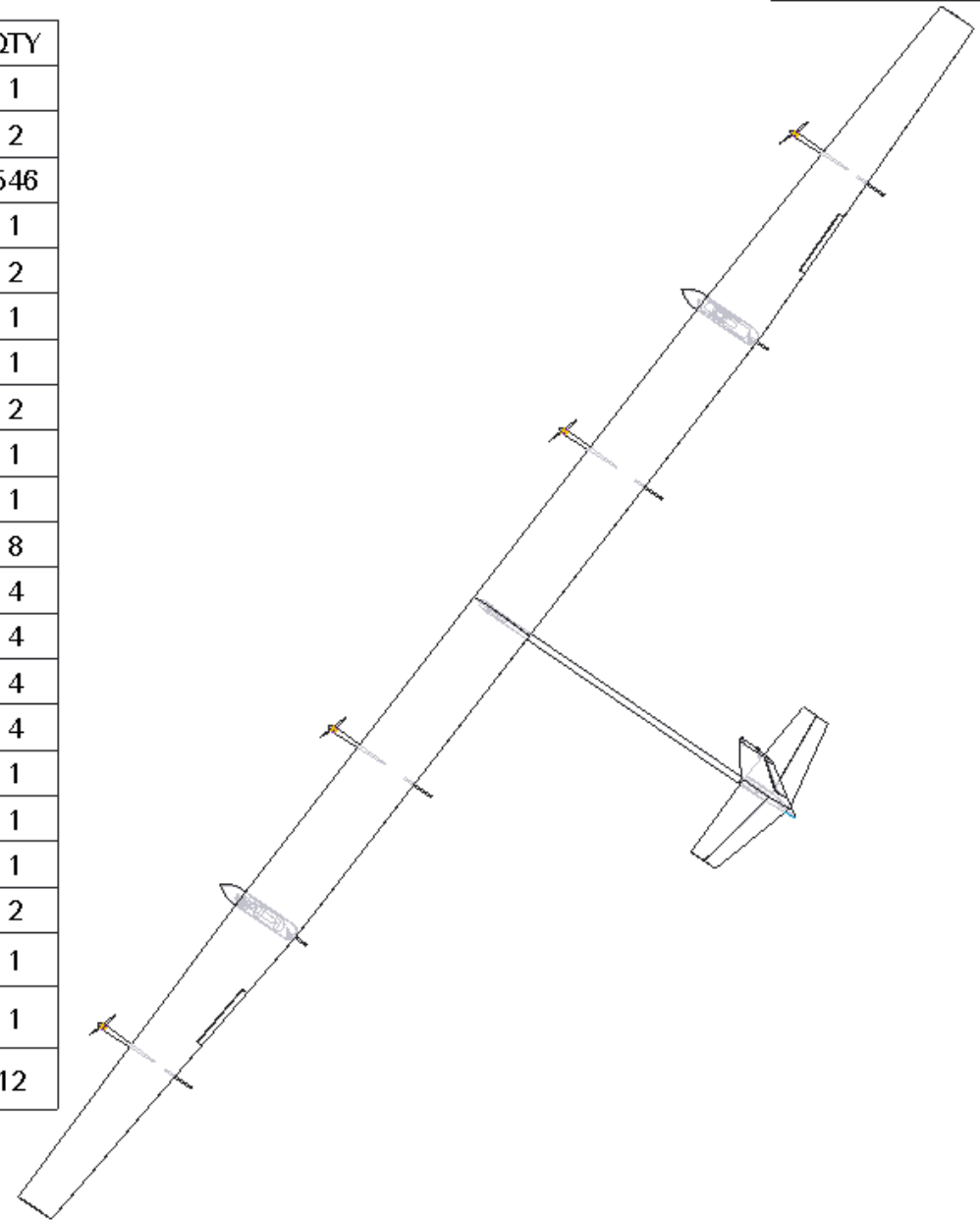
High-dynamic-range imaging (HDRI) used to generate these computer graphics.
These images were used for the report covers (upper left), poster (upper right) and presentations (lower).

Annex H: Technical Drawings

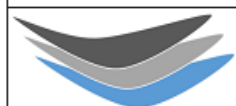
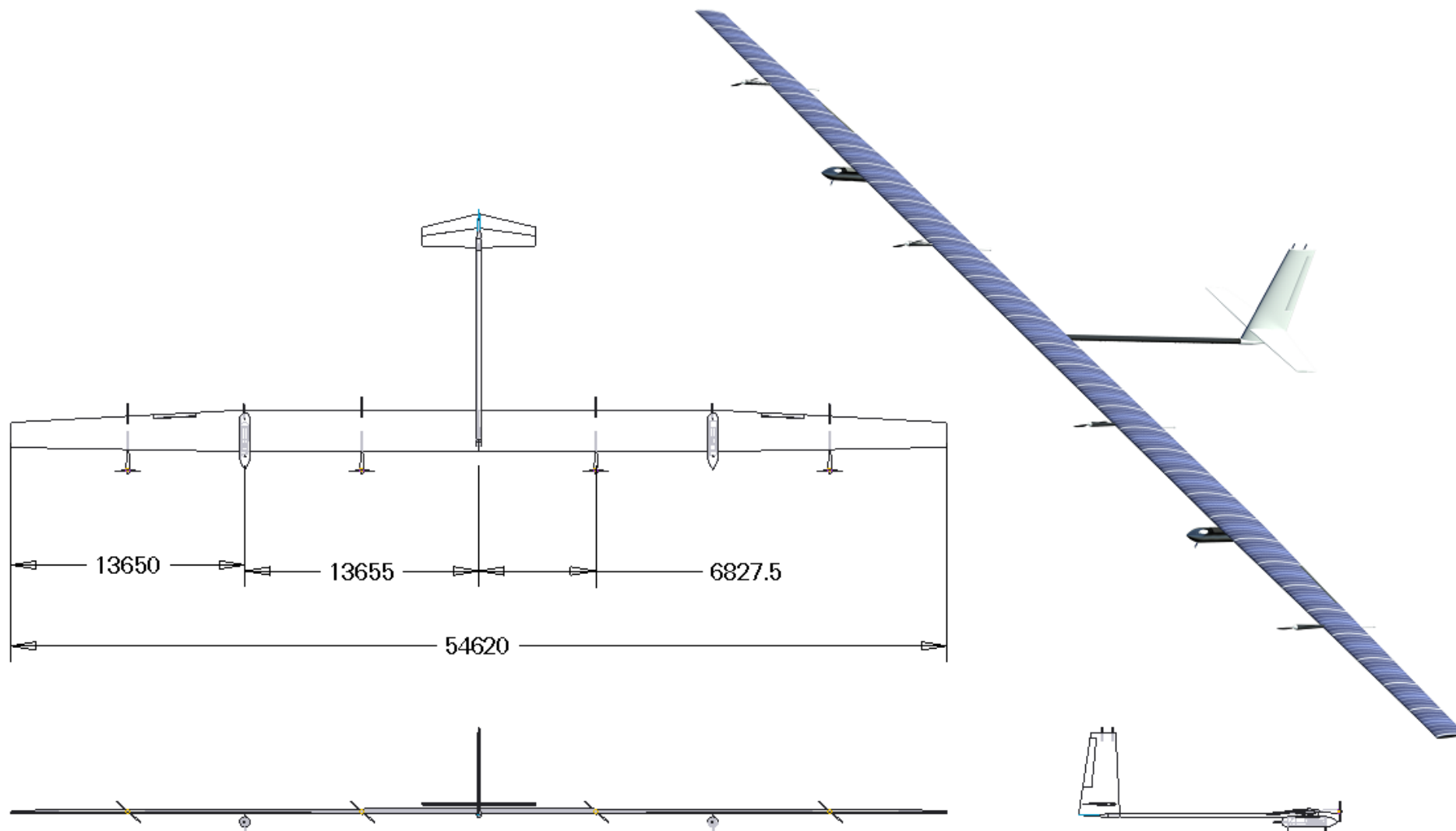
In order to preserve an accurate scale in printed-form, technical drawings will be reproduced on the next page.

- Intentionally left blank -

Label	Assembly	Sub Label	Sub Assembly	Materials	QTY
A	Main Wing	1	Skin	Ethylene Propylene	1
		2	Spar	CFRP	2
		3	Ribs	Spruce	546
		4	Solar Array	Various	1
		5	Aileron	CFRP	2
B	Tailplane	1	Vertical	Various	1
		2	Horizontal	Various	1
		3	Elevator	CFRP	2
		4	Rudder	CFRP	1
		5	Boom	CFRP	1
C	Engine	1	Blades	CFRP 0/90/0	8
		2	Cone	Ethylene Propylene	4
		3	Motor	Various (Metals)	4
		4	Shaft	CFRP	4
		5	Skin	Ethylene Propylene	4
D	Pods	1	Fuels Cells	Various	1
		2	H2 Tank	Kevlar + Al lining	1
		3	O2 Tank	Kevlar + Al lining	1
		4	Water Tanks	Rubber elastomer	2
		5	Compressor	Various	1
E	Payload Module	1	Box	CFRP	1
		2	Antenna + Mount	Various	12



	Iliev, Sergiu	GDP Project	Global Overview	Tolerances: Refer to parts	Scale	All dimensions in mm	
	Jackson, Sam	Date: 12.6.2015	All Parts Shown	Finish: Refer to parts	0.0042		Drawing no.
Vehicle Definition	Wilson, Alex	Version: 3.3	Materials: Refer to parts drawings				



Vehicle Definition

Iliev, Sergiu

Jackson, Sam

Wilson, Alex

GDP Project

Date: 12.6.2015

Version: 3.3

Global 3-view

All Parts Shown

Materials: Refer to parts drawings

Tolerances: Refer to parts

Finish: Refer to parts

Scale

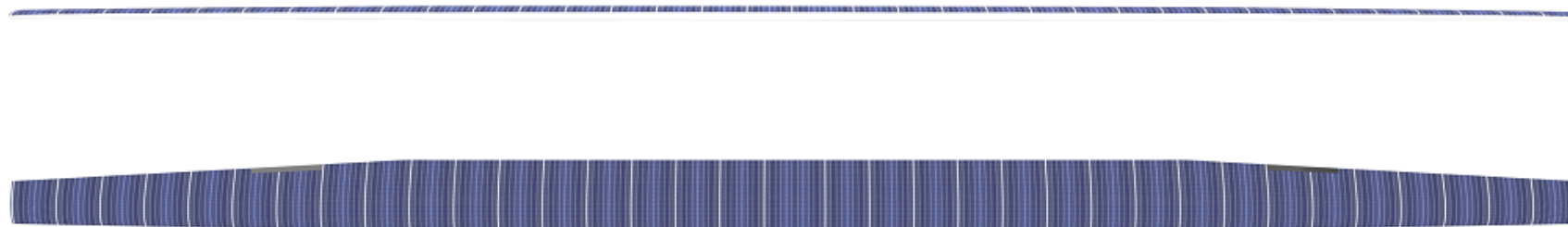
0.001

All dimensions in mm

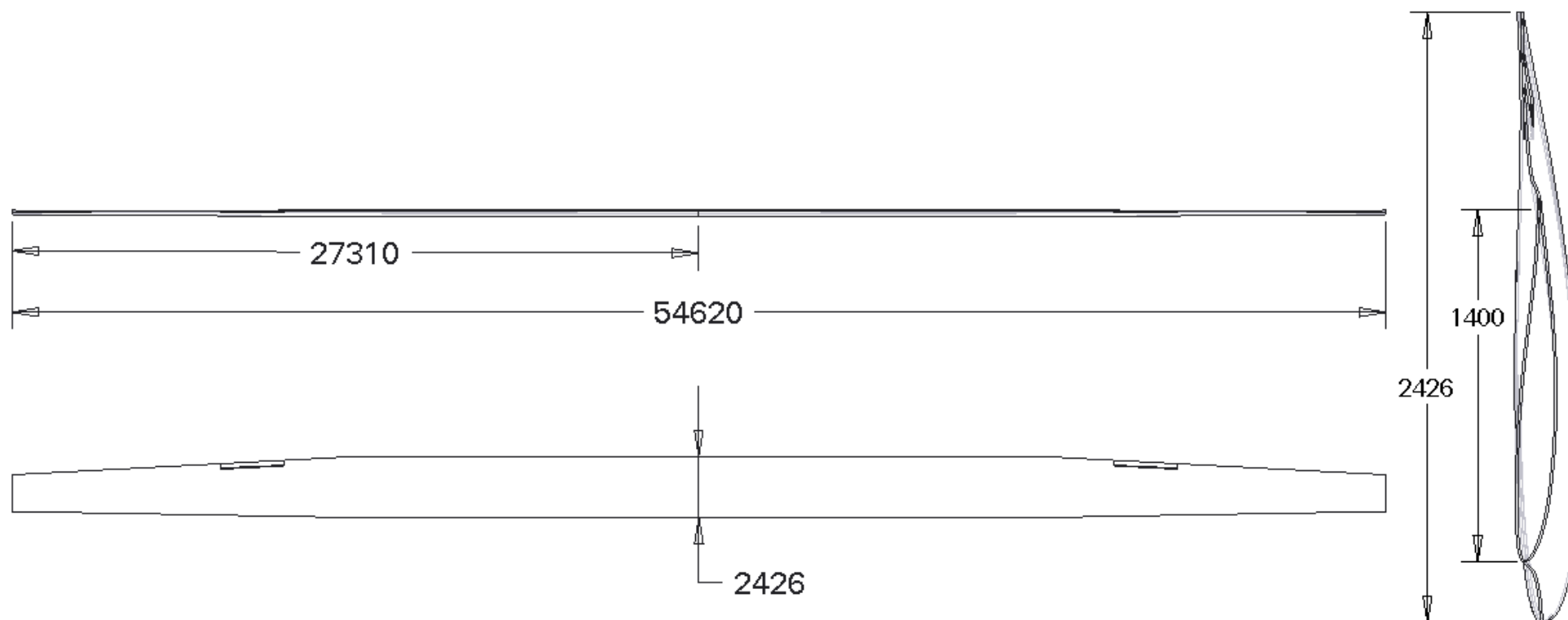


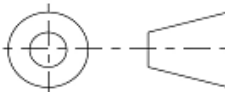
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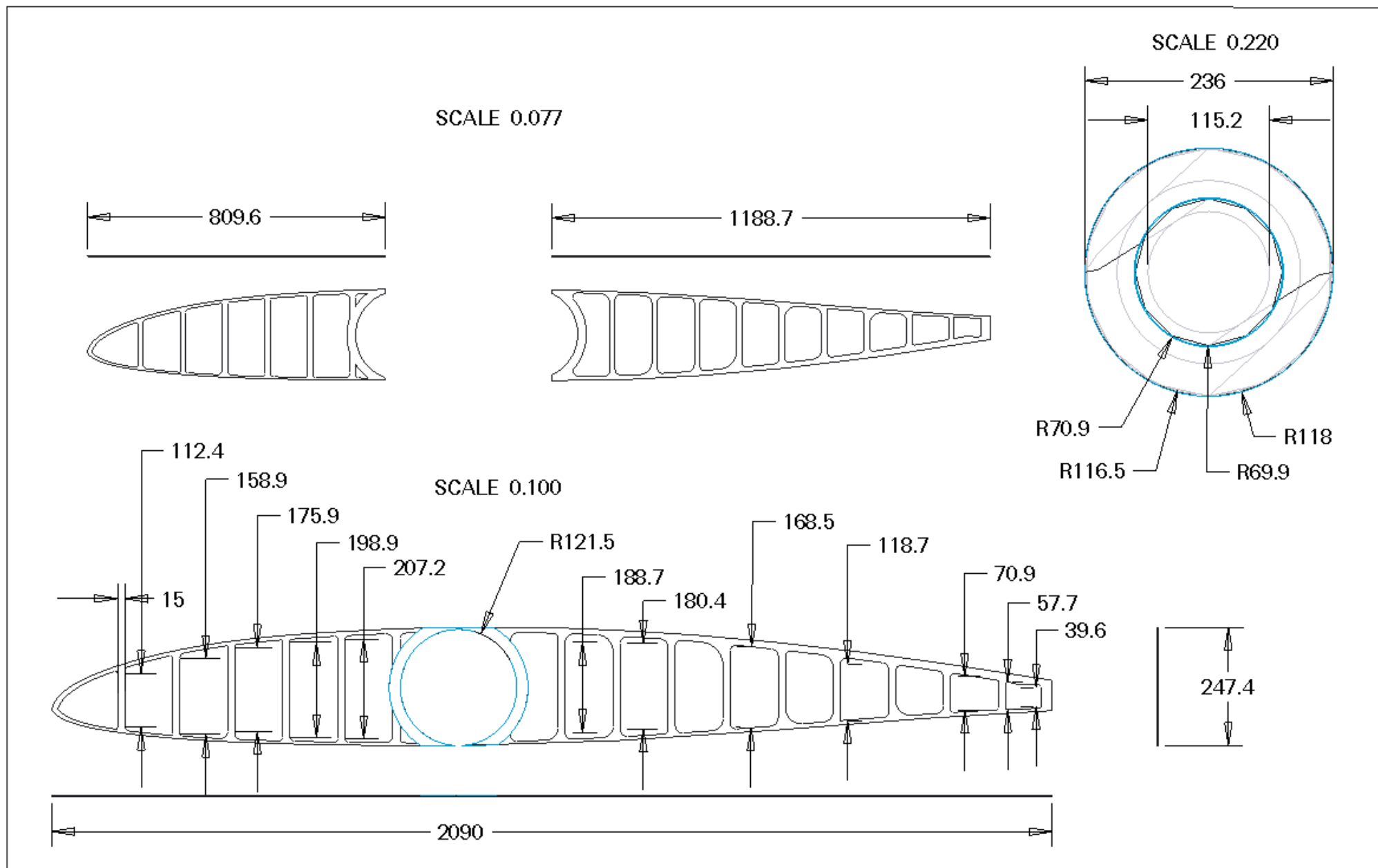
2/19

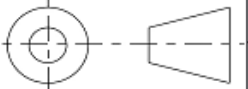


Section
SCALE 0.045

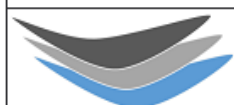
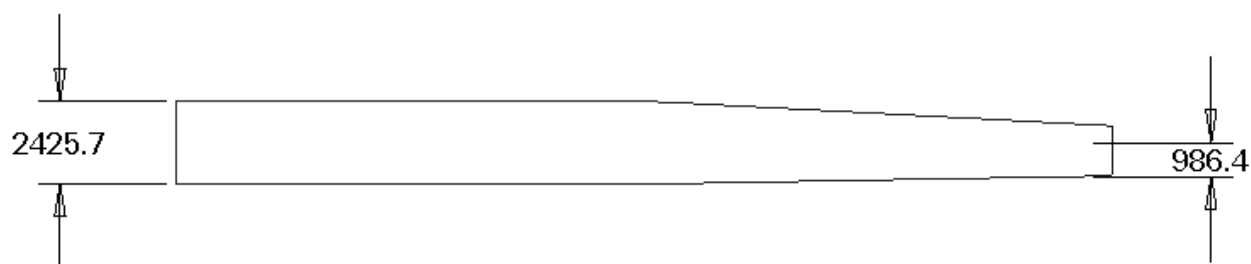
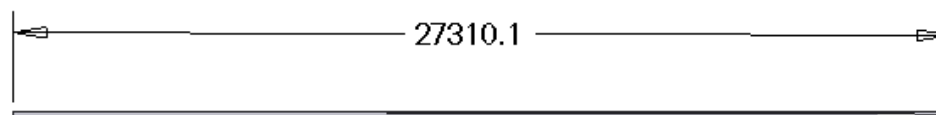
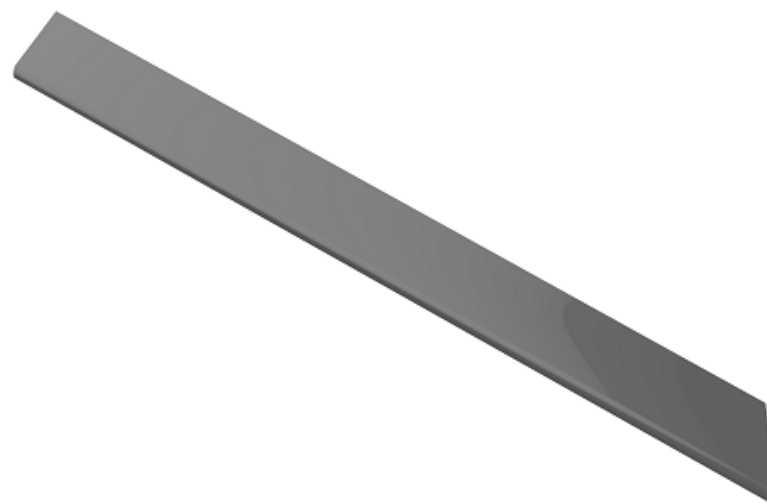
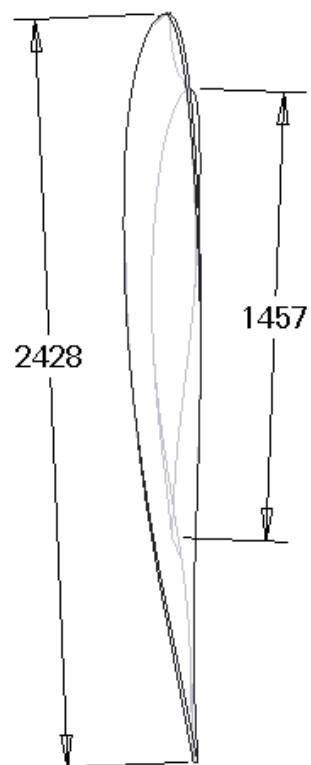


	Iliev, Sergiu	GDP Project	Main Wing	Tolerances: ± 0.1 mm	Scale	All dimensions in mm	
	Jackson, Sam	Date: 12.6.2015	A 1-5	Finish: Refer to parts			Drawing no.
	Wilson, Alex	Version: 3.3	Materials: Refer to parts drawings		0.0045		3/19



 Vehicle Definition	Iliev, Sergiu	GDP Project	Wing Section Parts	Tolerances: ± 0.1 mm	Scale	All dimensions in mm	
	Jackson, Sam	Date: 12.6.2015	A 1-4	Finish: Refer to parts	Various		Drawing no.
	Wilson, Alex	Version: 3.3	Materials: CFRP for spar, Spruce for Ribs				4/19

SCALE 0.045



Vehicle Definition

Iliev, Sergiu

Jackson, Sam

Wilson, Alex

GDP Project

Date: 12.6.2015

Version: 3.3

Aileron

A 5

Materials: CFRP

Tolerances: ± 0.1 mm

Finish: Epoxy UV resin

Scale

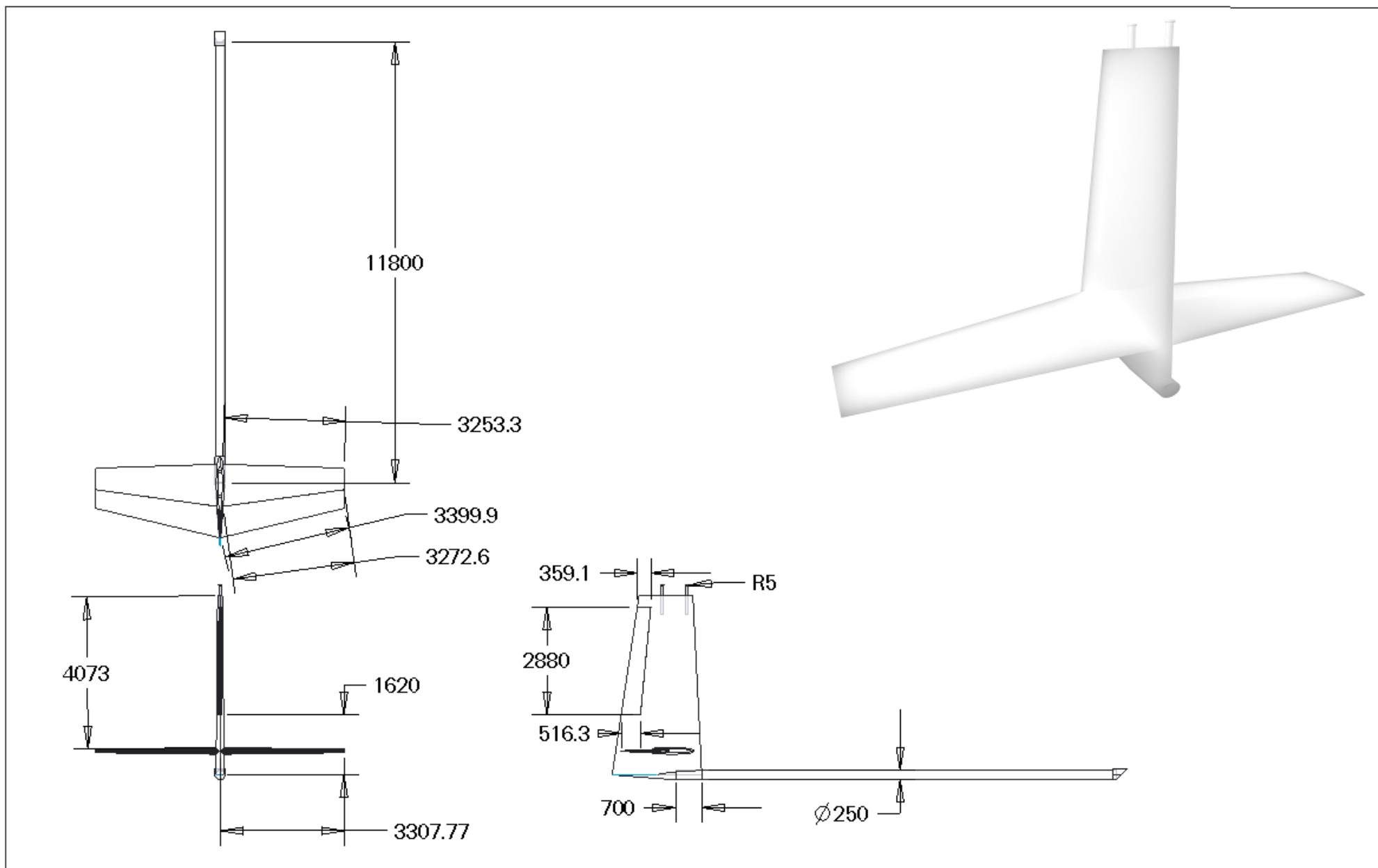
0.0045

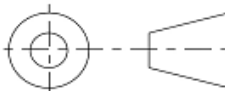
All dimensions in mm

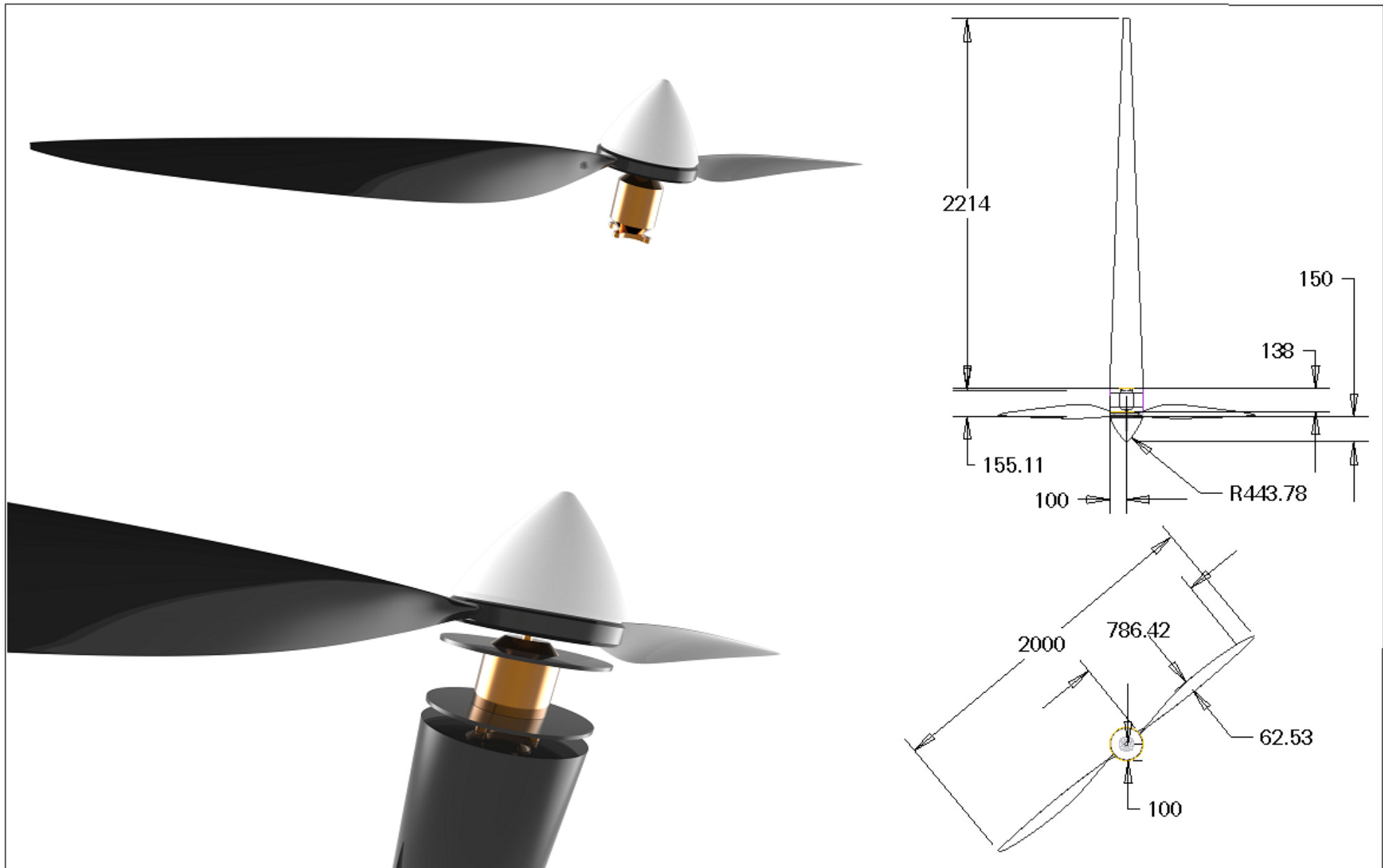


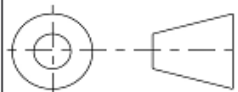
Drawing no.

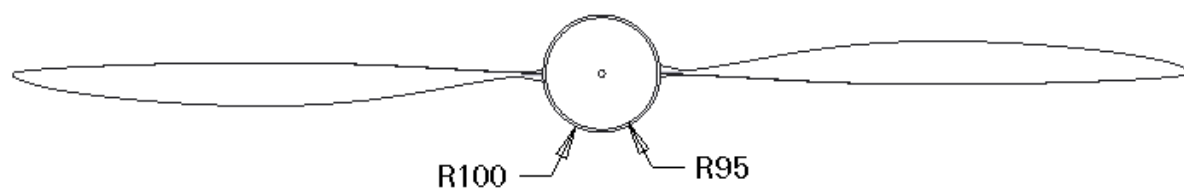
5/19



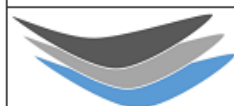
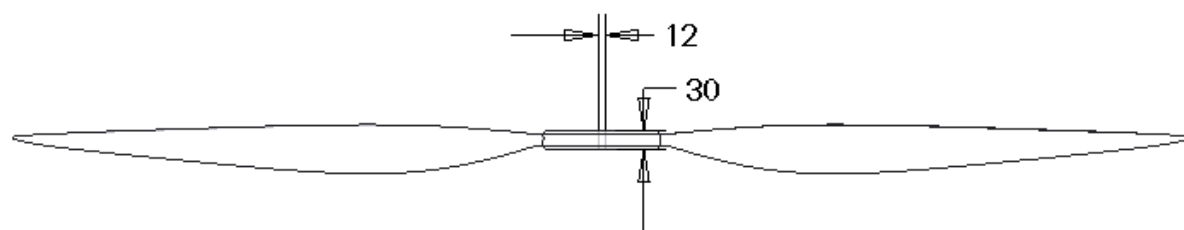
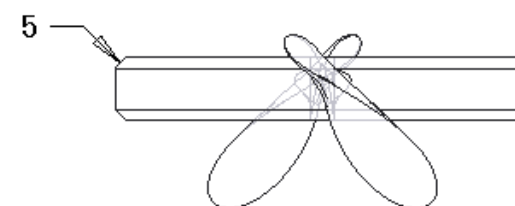
 Vehicle Definition	Iliev, Sergiu	GDP Project	Tailplane	Tolerances: ± 1 mm	Scale	All dimensions in mm	
	Jackson, Sam	Date: 12.6.2015	B 1-5	Finish: Refer to parts	0.008		Drawing no.
	Wilson, Alex	Version: 3.3	Materials: CFRP Boom, EP skin, Spruce ribs				6/19



	Iliev, Sergiu	GDP Project	Engines	Tolerances: ± 0.01 mm	Scale	All dimensions in mm	
	Jackson, Sam	Date: 12.6.2015	C 1-5	Finish: Refer to parts	0.035		Drawing no.
Vehicle Definition	Wilson, Alex	Version: 3.3	Materials: Refer to parts drawings				



SCALE 0.300



Iliev, Sergiu

GDP Project

Blade

Tolerances: ± 0.01 mm

Scale

All dimensions in mm

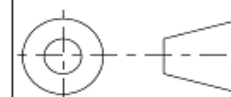
Jackson, Sam

Date: 12.6.2015

C1

Finish: Smooth

0.085



Drawing no.

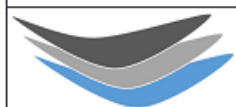
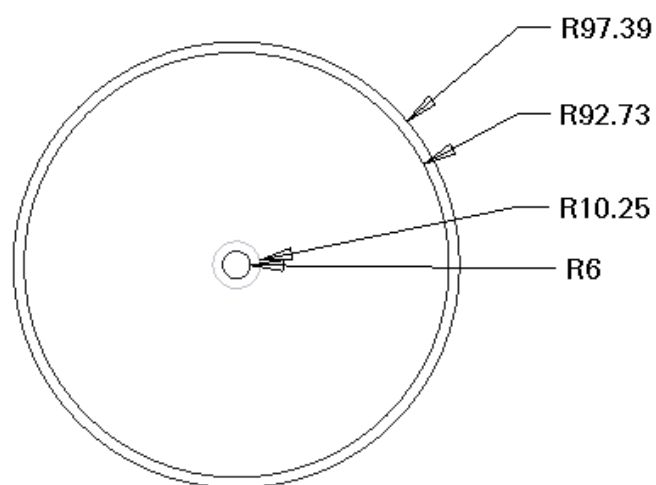
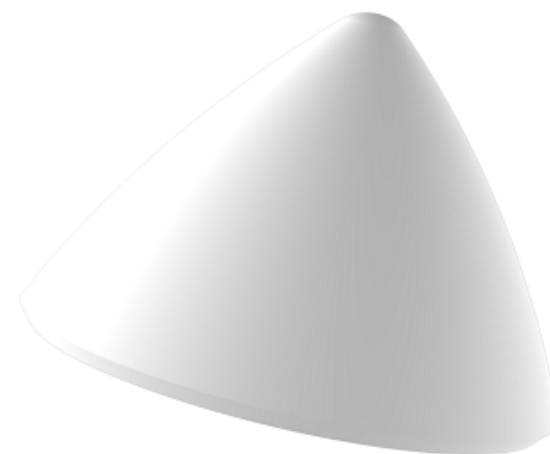
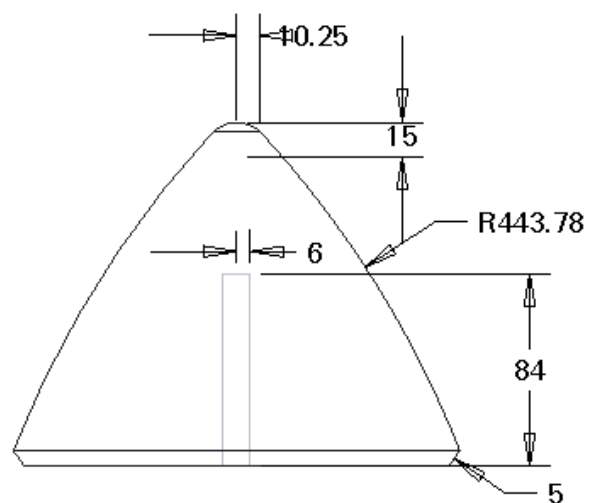
Vehicle Definition

Wilson, Alex

Version: 3.3

Materials: CFRP

8/19



Vehicle Definition

Iliev, Sergiu

Jackson, Sam

Wilson, Alex

GDP Project

Date: 12.6.2015

Version: 3.3

Cone

C2

Materials: Ethylene Propylene

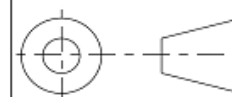
Tolerances: ± 0.01 mm

Finish: Smooth

Scale

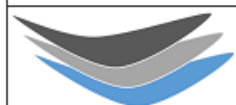
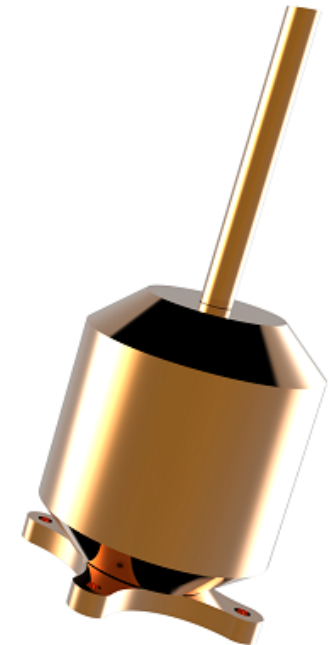
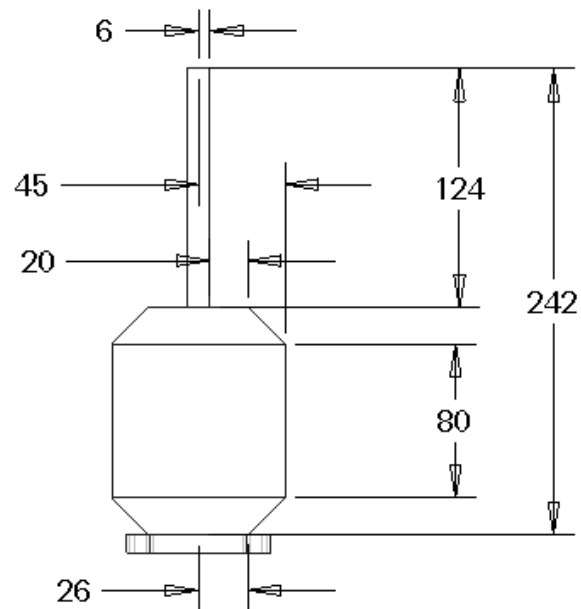
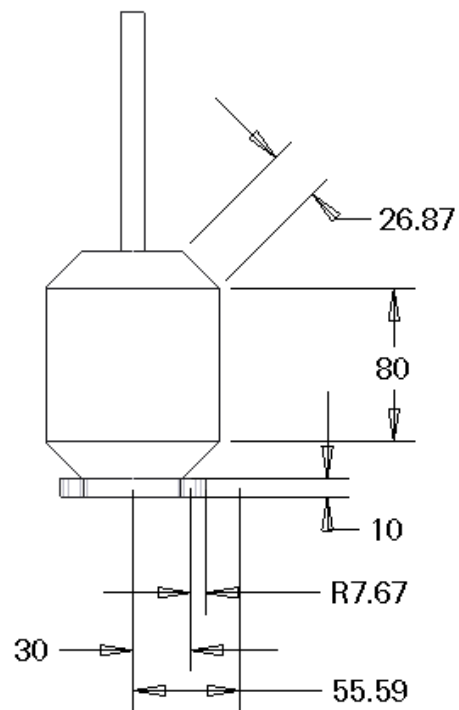
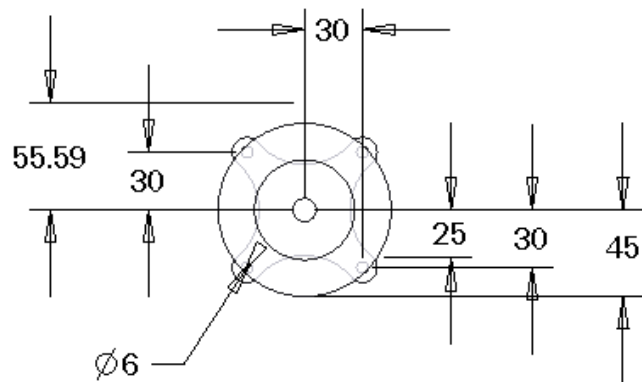
0.333

All dimensions in mm



Drawing no.

9/19



Vehicle Definition

Iliev, Sergiu

Jackson, Sam

Wilson, Alex

GDP Project

Date: 12.6.2015

Version: 3.3

Motor

C 3

Materials: Various

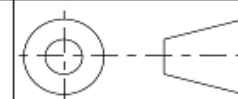
Tolerances: ± 0.01 mm

Finish: Smooth

Scale

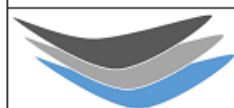
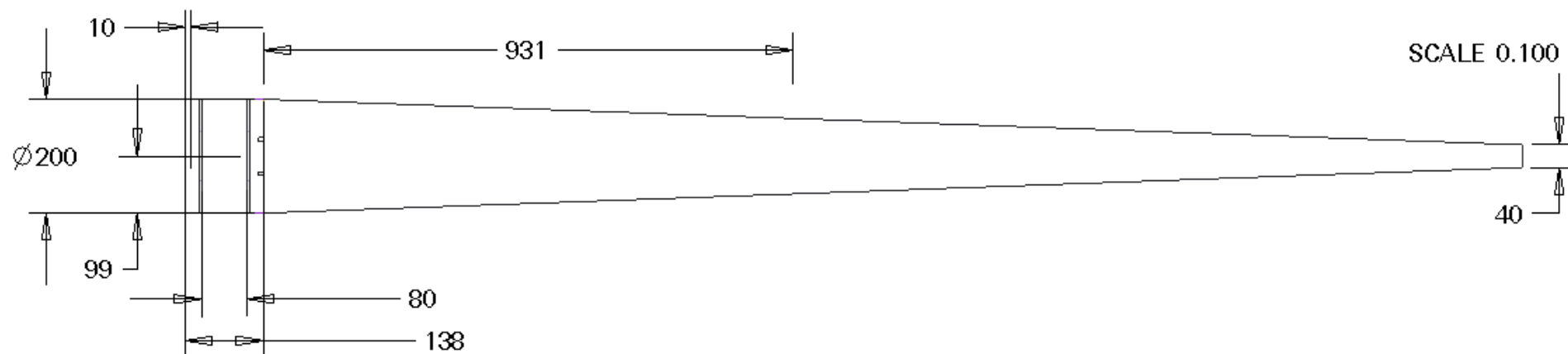
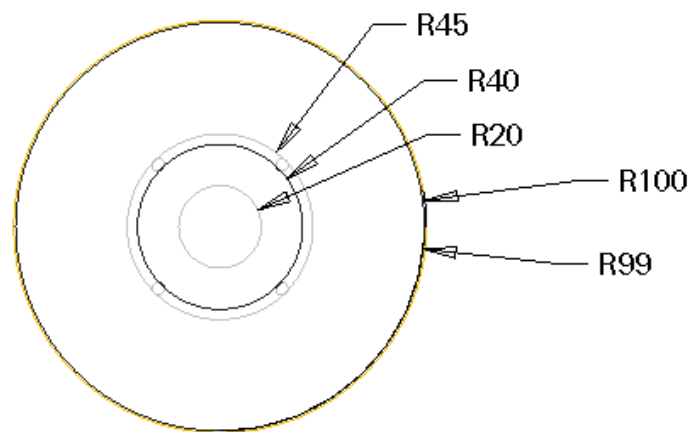
0.280

All dimensions in mm



Drawing no.

10/19



Vehicle Definition

Iliev, Sergiu

Jackson, Sam

Wilson, Alex

GDP Project

Date: 12.6.2015

Version: 3.3

Shaft & Skin

C 4-5

Materials: Shaft is CFRP, Wrap is EP

Tolerances: ± 0.1 mm

Finish: Refer to parts

Scale

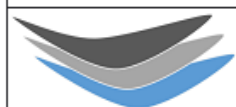
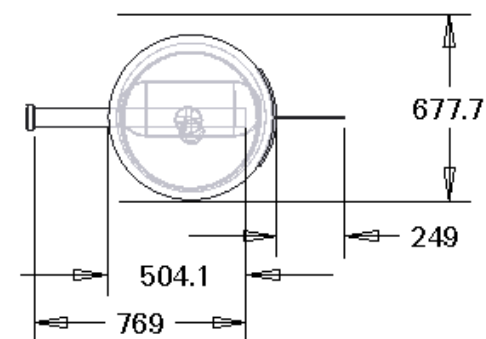
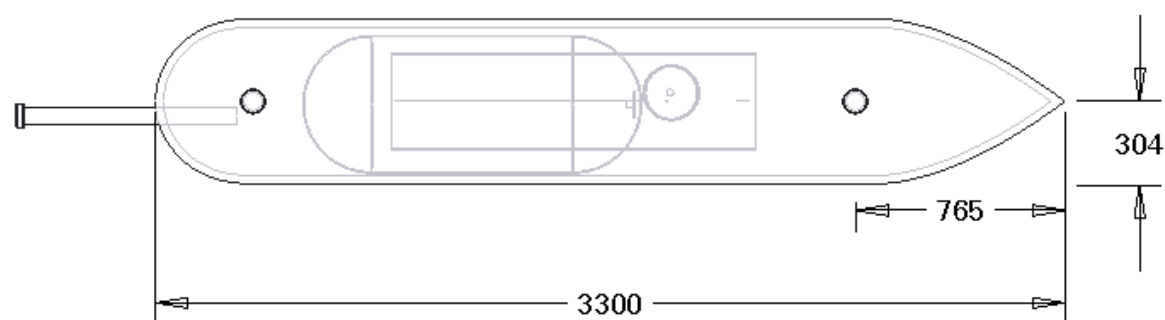
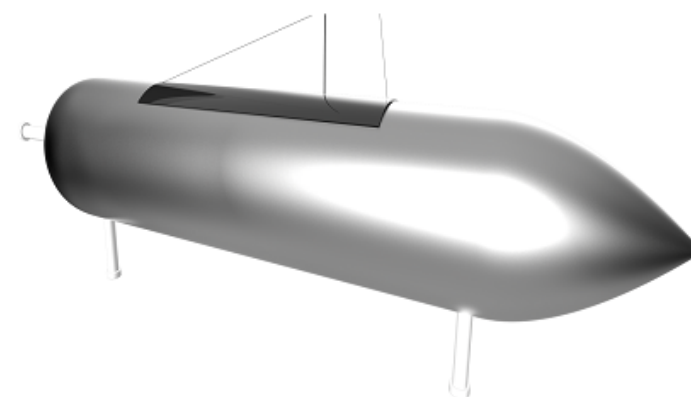
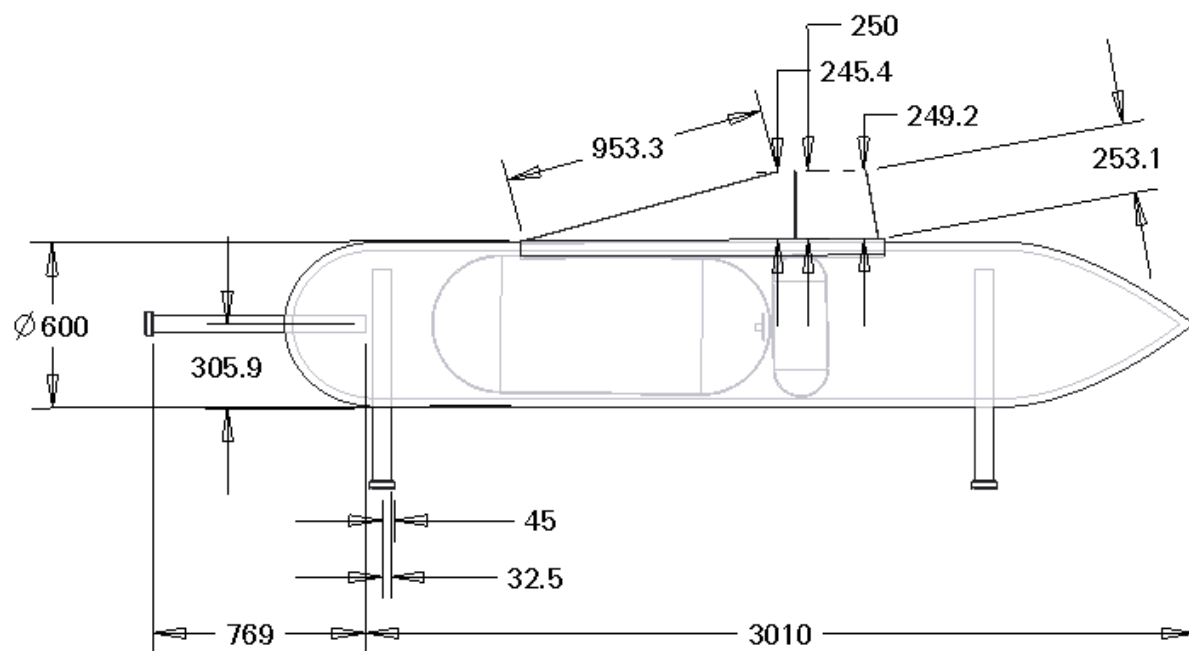
0.300

All dimensions in mm



Drawing no.

11/19



Vehicle Definition

Iliev, Sergiu

Jackson, Sam

Wilson, Alex

GDP Project

Date: 12.6.2015

Version: 3.3

Left Pod

D 2-3

Materials: Refer to parts drawings

Tolerances: ± 0.1 mm

Finish: Refer to parts

Scale

0.040

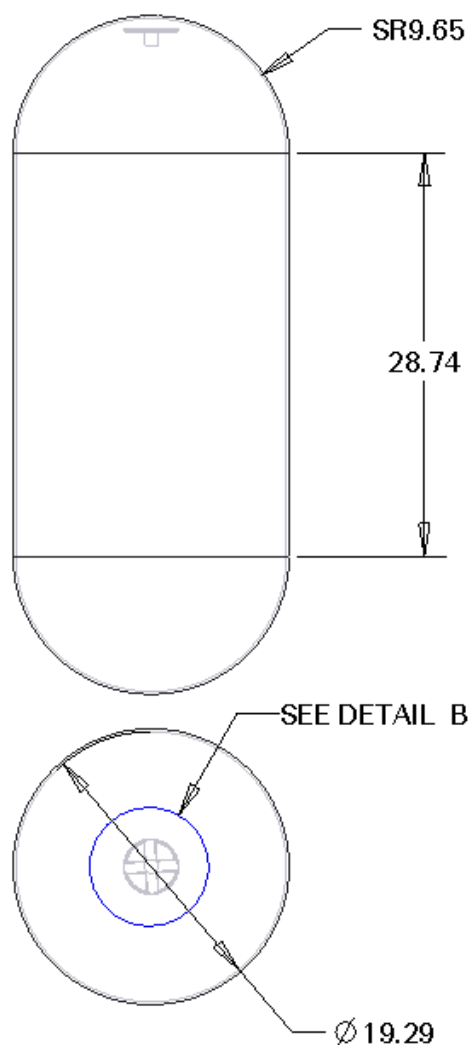
All dimensions in mm



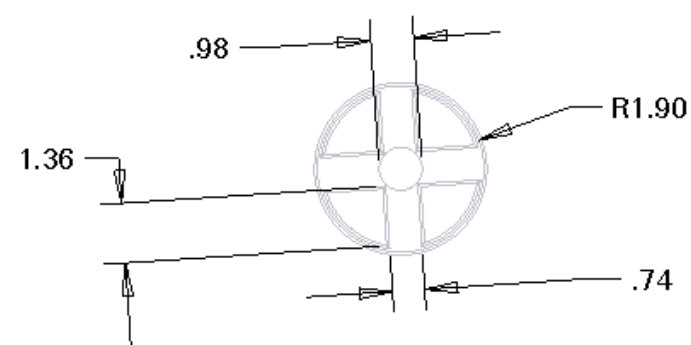
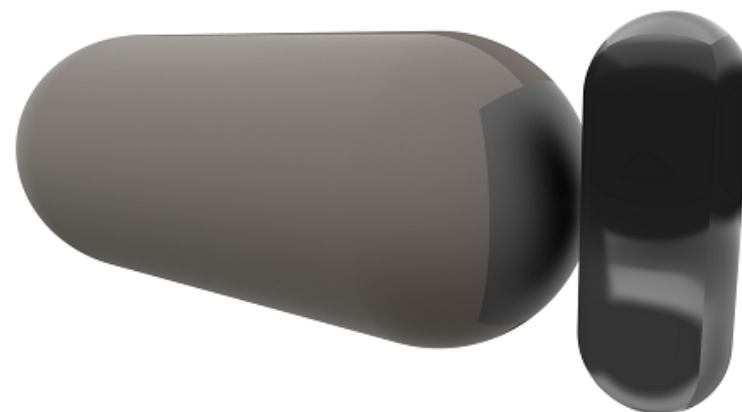
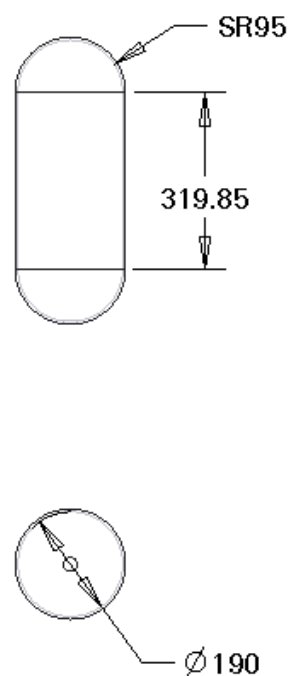
Drawing no.

12/19

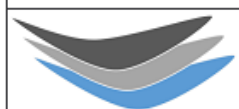
Hydrogen Tank



Oxygen Tank



DETAIL B
SCALE 0.250



Vehicle Definition

Iliev, Sergiu

Jackson, Sam

Wilson, Alex

GDP Project

Date: 12.6.2015

Version: 3.3

H2 & O2 tanks

D 2-3

Materials: Kevlar + Al lining

Tolerances: ± 0.01 mm

Finish: fabric texture

Scale

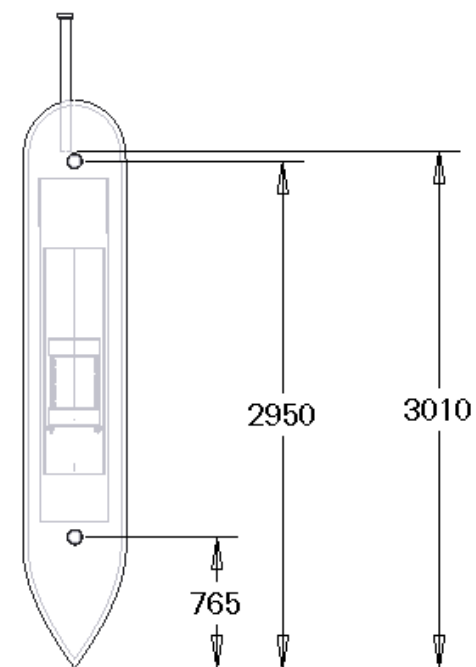
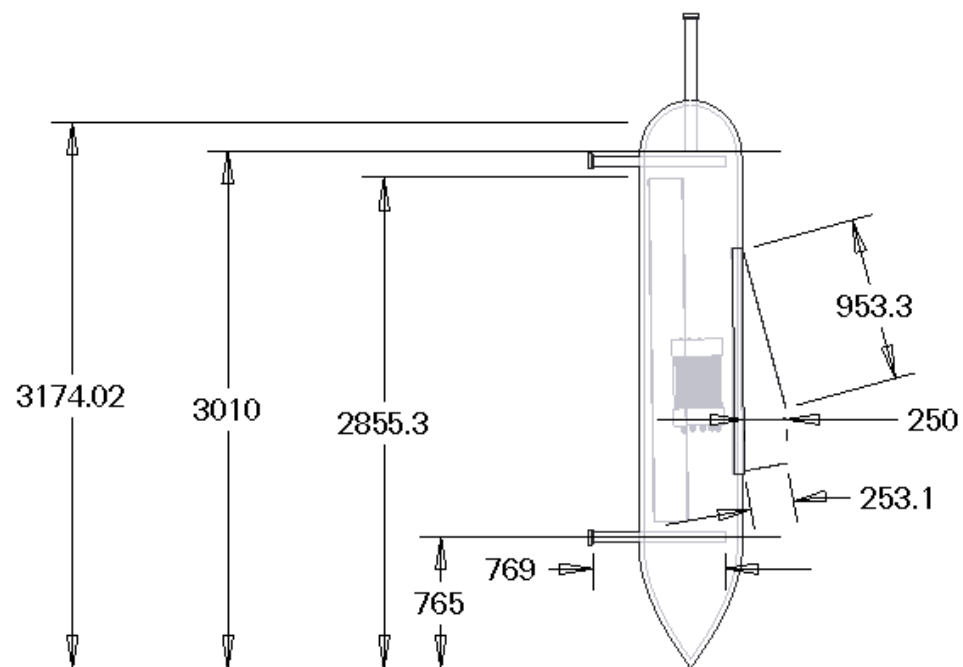
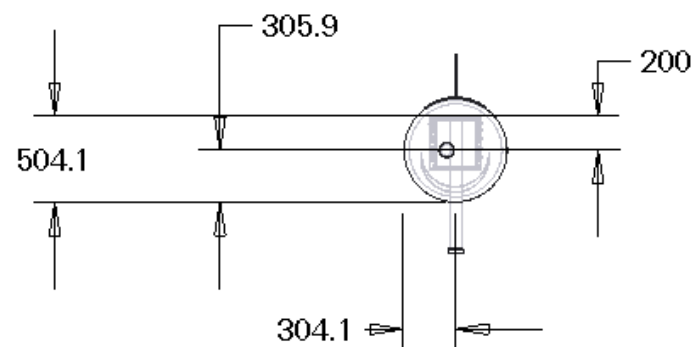
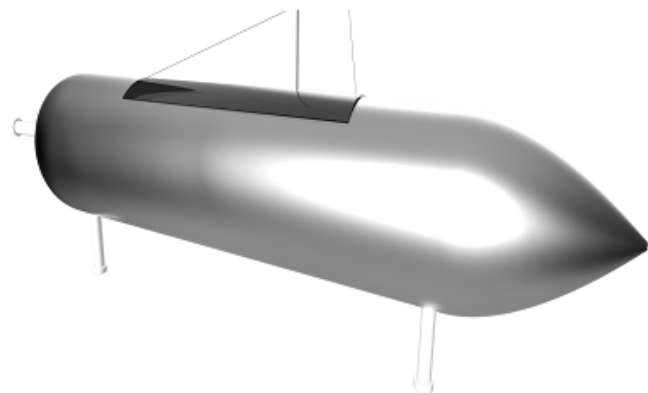
0.080

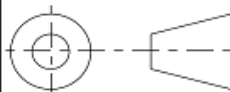
All dimensions in mm

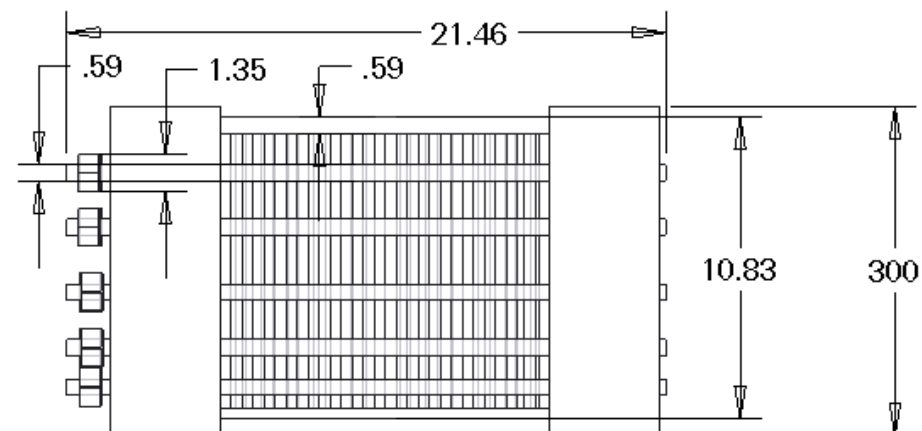
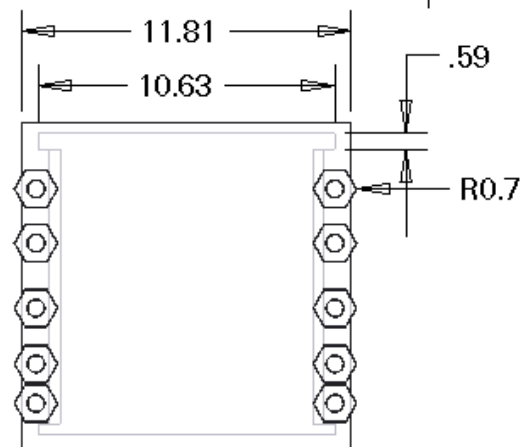
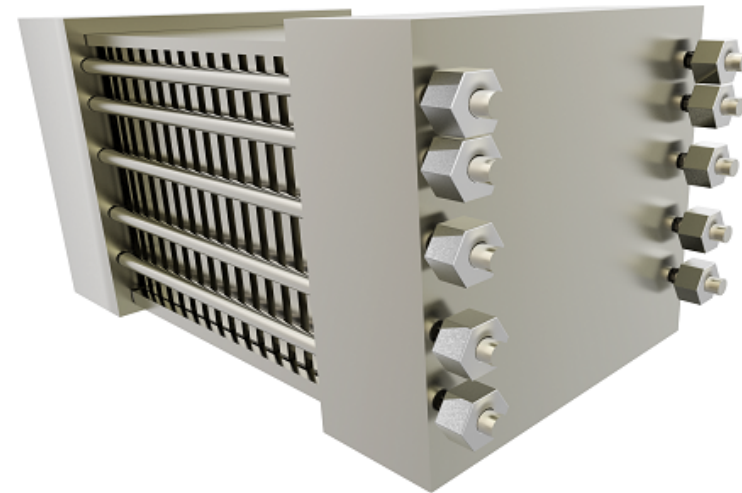
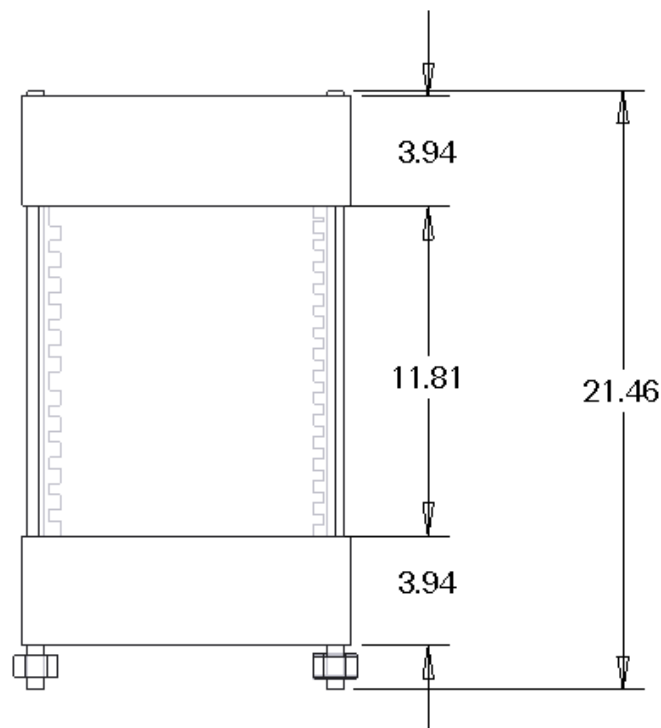


Drawing no.

13/19



	Iliev, Sergiu	GDP Project	Right Pod	Tolerances: ± 0.1 mm	Scale	All dimensions in mm	
	Jackson, Sam	Date: 12.6.2015	D 1, 4-5	Finish: Refer to parts	0.025		Drawing no.
	Wilson, Alex	Version: 3.3	Materials: Refer to parts drawings				14/19
Vehicle Definition							



Vehicle Definition

Iliev, Sergiu

Jackson, Sam

Wilson, Alex

GDP Project

Date: 12.6.2015

Version: 3.3

Fuel Cell

E1

Materials: Various, Pt catalyst, Al shell

Tolerances: ± 0.01 mm

Finish: Polished

Scale

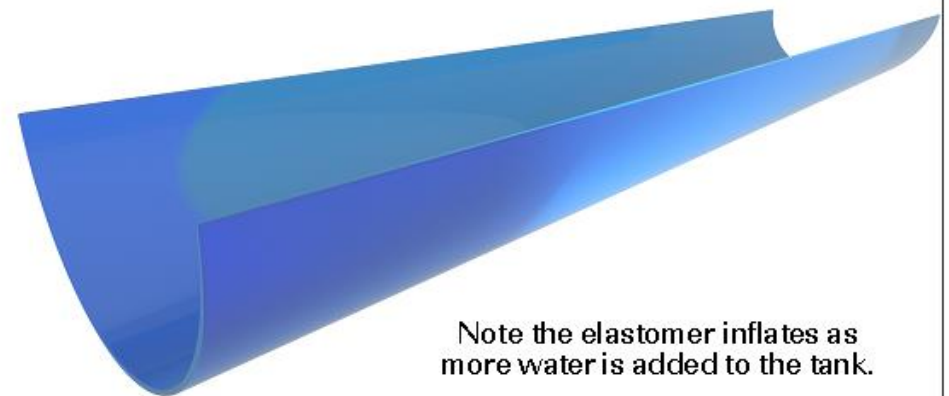
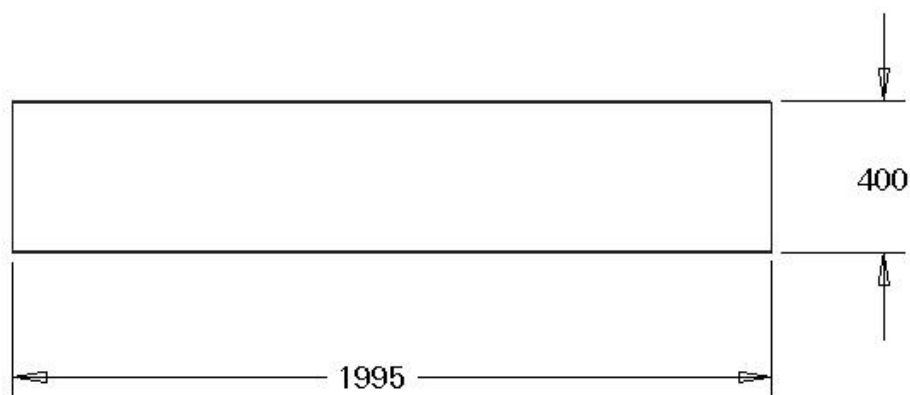
0.160

All dimensions in mm

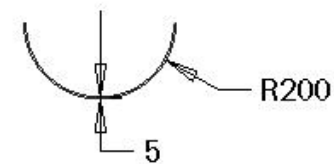
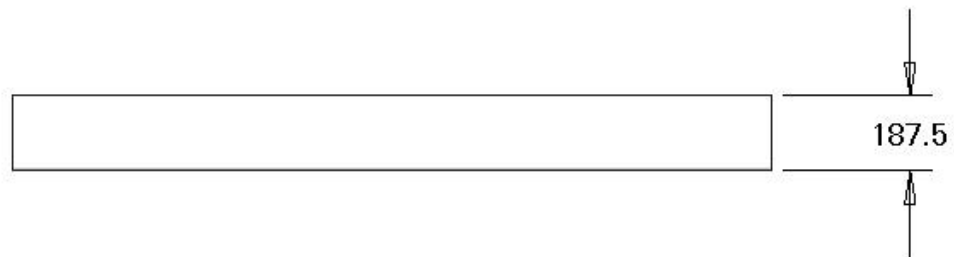


Drawing no.

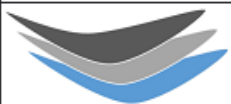
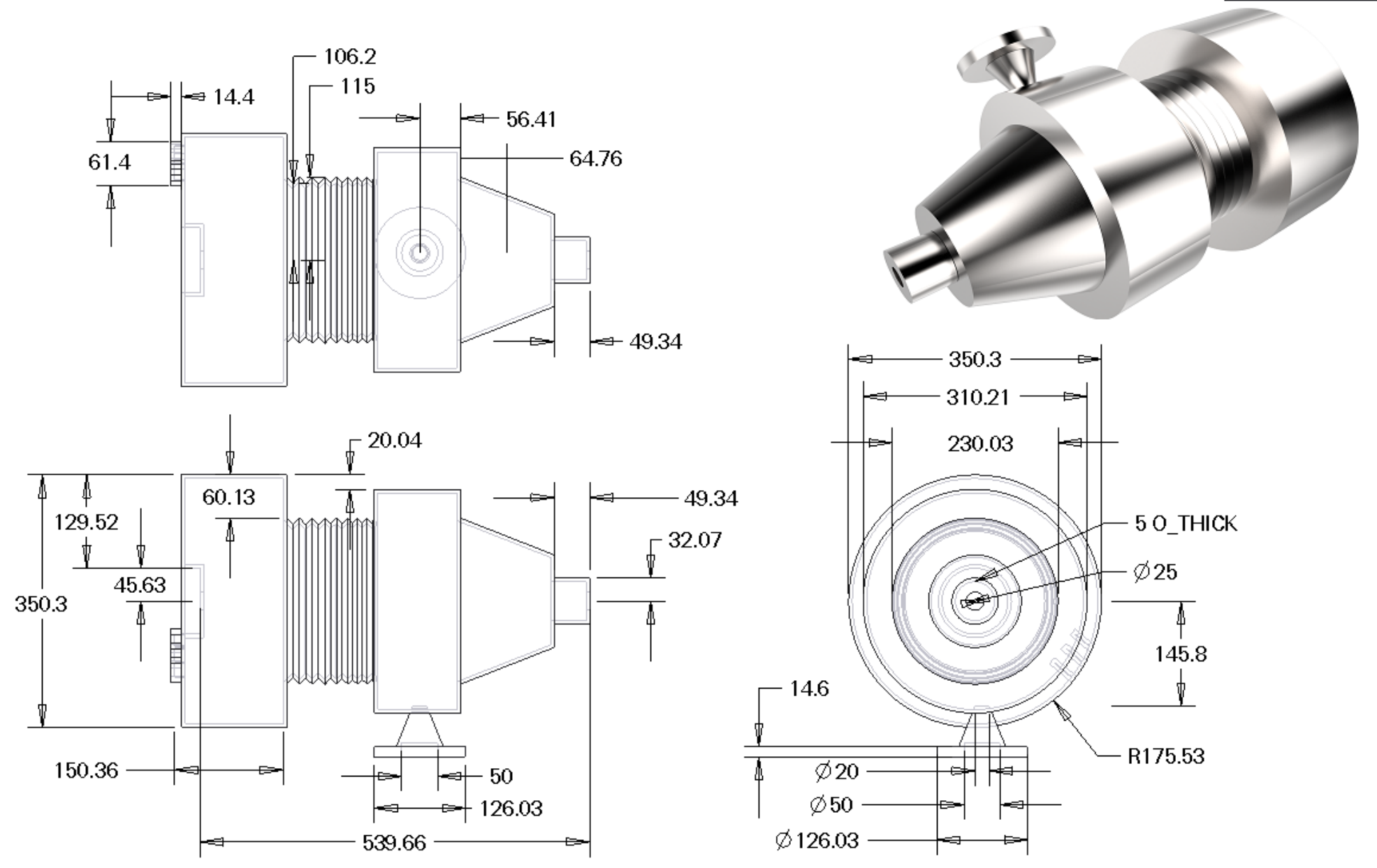
15/19



Note the elastomer inflates as more water is added to the tank.



	Iliev, Sergiu	GDP Project	Water tank	Tolerances: ± 0.1 mm	Scale	All dimensions in mm	
	Jackson, Sam	Date: 12.6.2015	D4	Finish: Smooth	0.055		Drawing no.
	Vehicle Definition	Wilson, Alex	Version: 3.3	Materials: Rubberized elastomer			16/19



Vehicle Definition

Iliev, Sergiu

Jackson, Sam

Wilson, Alex

GDP Project

Date: 12.6.2015

Version: 3.3

Compressor

D5

Materials: Aluminium Alloy

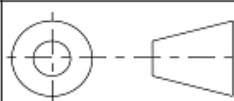
Tolerances: ± 0.01 mm

Finish: Milled finish

Scale

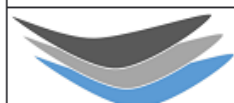
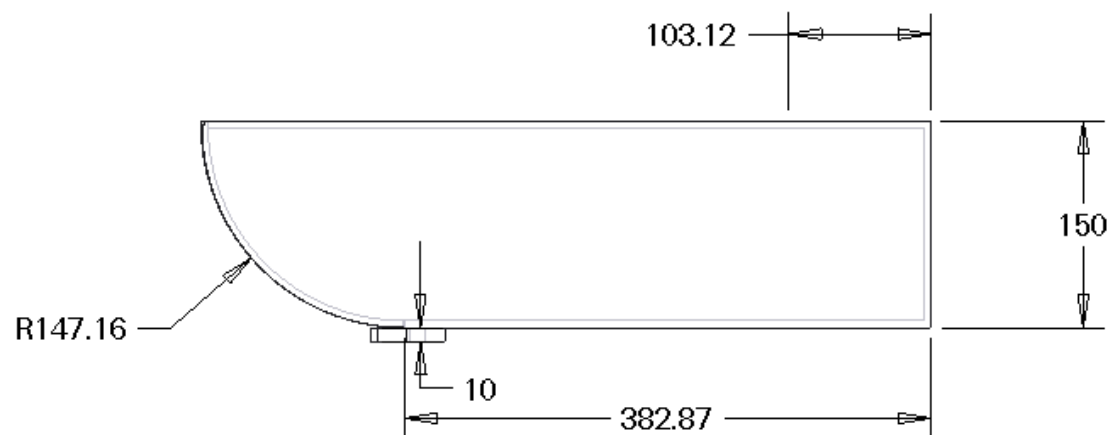
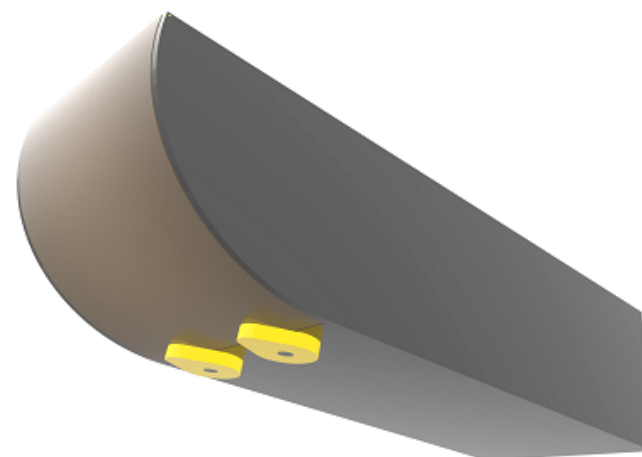
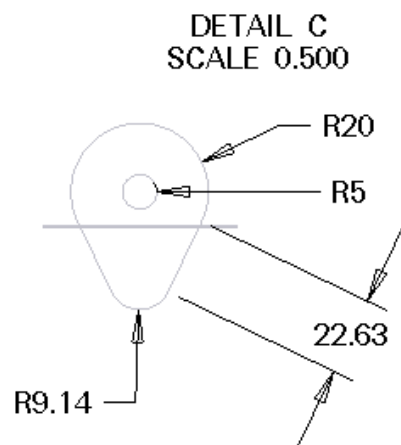
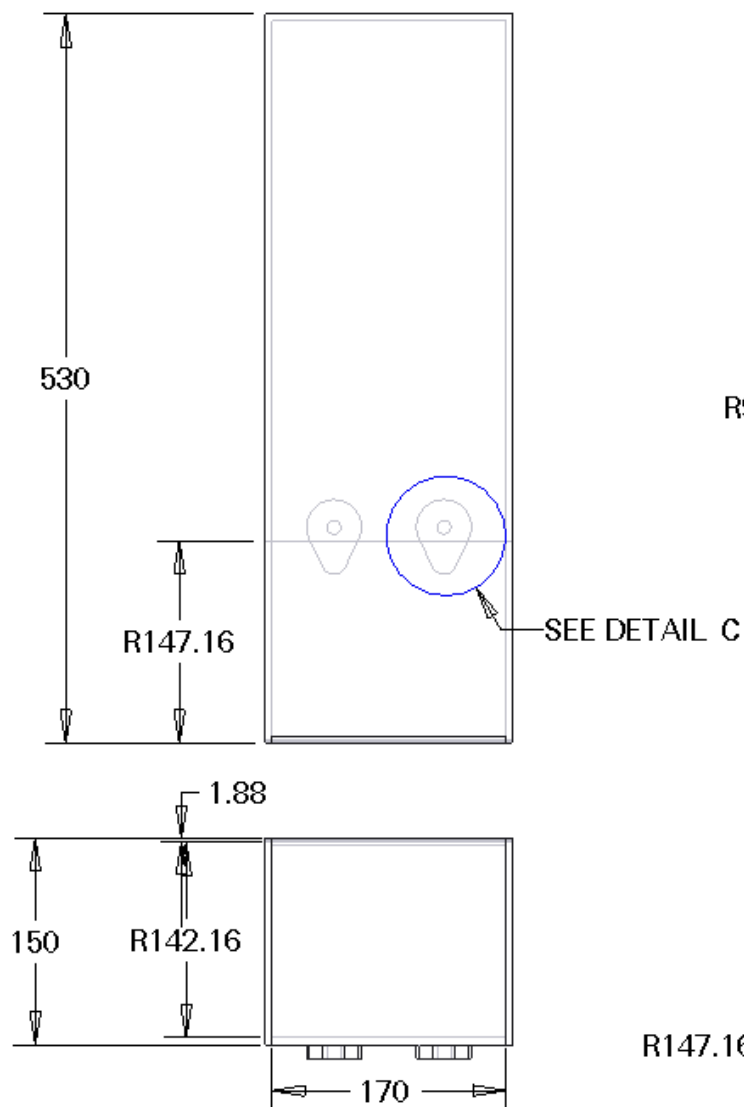
0.150

All dimensions in mm



Drawing no.

17/19



Vehicle Definition

Iliev, Sergiu

Jackson, Sam

Wilson, Alex

GDP Project

Date: 12.6.2015

Version: 3.3

Payloads

E1

Materials: CFRP body GFRP front

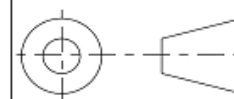
Tolerances: ± 0.01 mm

Finish: Smooth

Scale

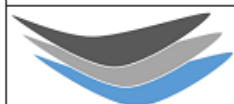
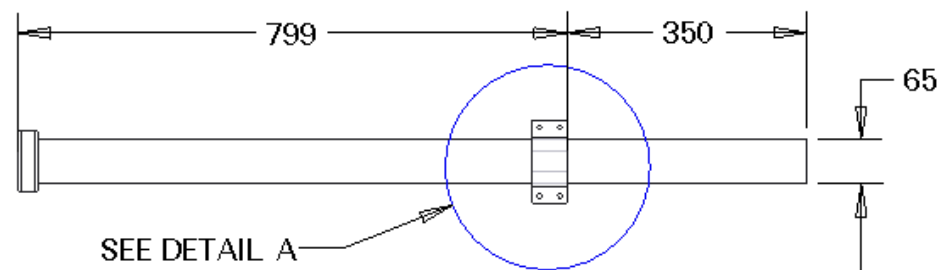
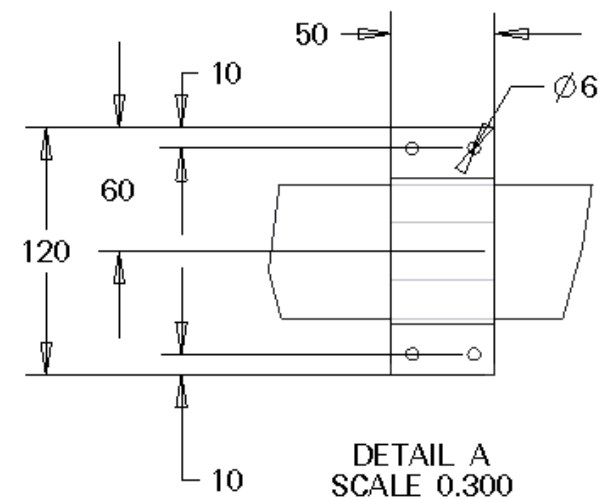
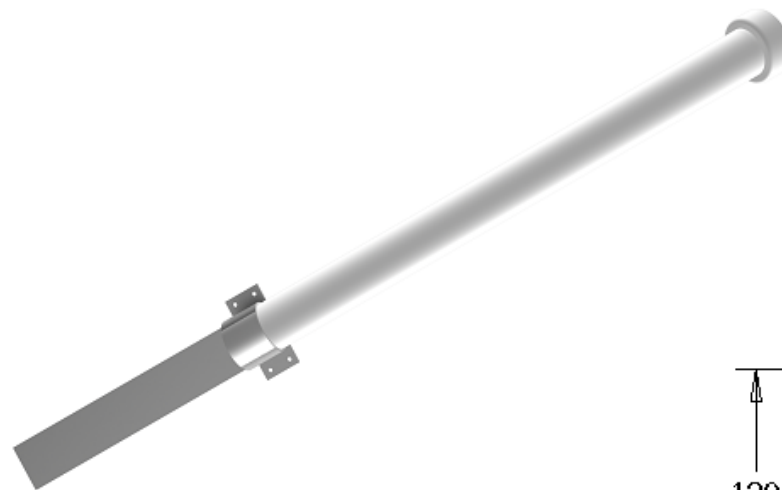
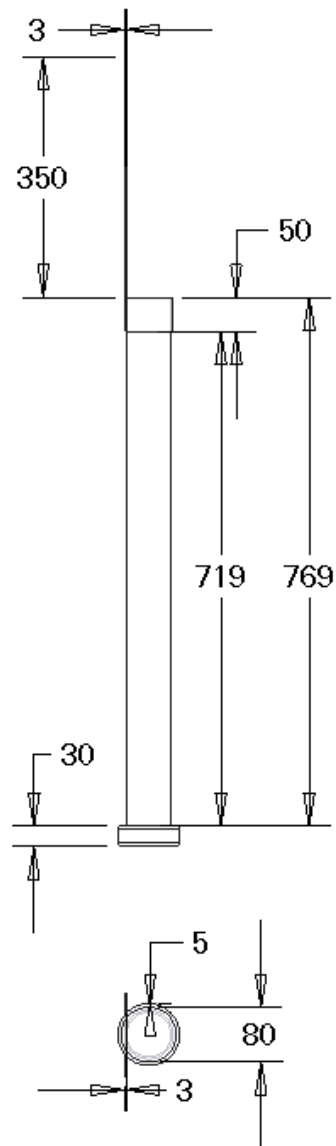
0.200

All dimensions in mm



Drawing no.

18/19



Vehicle Definition

Iliev, Sergiu

Jackson, Sam

Wilson, Alex

GDP Project

Date: 12.6.2015

Version: 3.3

Antenna

E2

Materials: Al fo holder, Various for antenna

Tolerances: ± 0.5 mm

Finish of holder: smooth

Scale

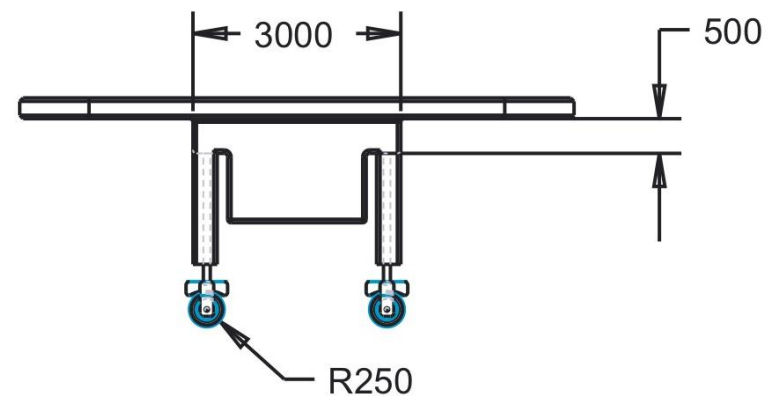
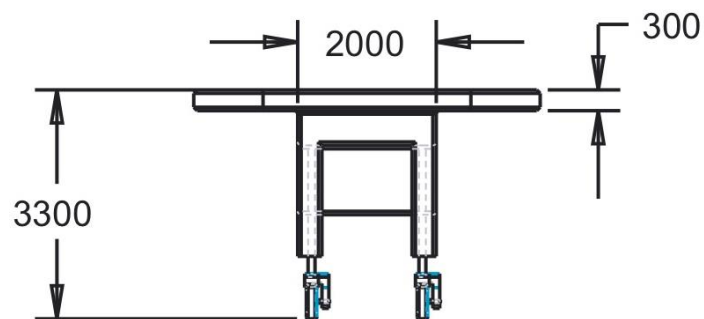
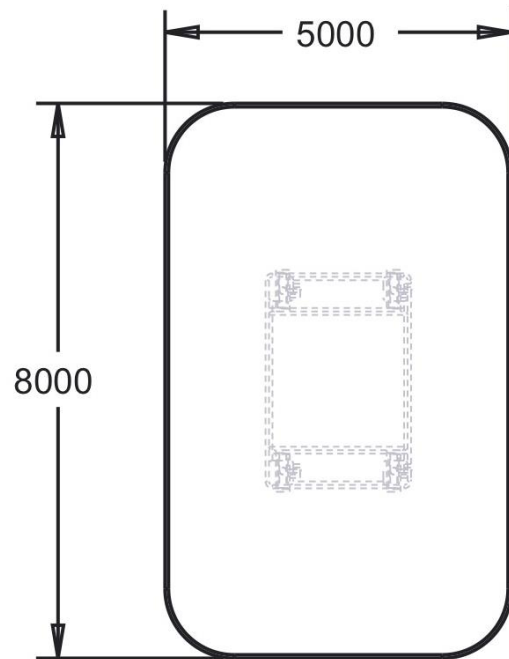
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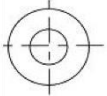
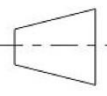
All dimensions in mm



Drawing no.

19/19



	Iliev, Sergiu	GDP Project	M.A.R.V.	Tolerances: Specific to individual parts	Scale	All dimensions in mm	
	Jackson, Sam	Date: 12.6.2015		Finish: Specific to individual parts	0.010		Drawing no.
	Wilson, Alex	Version: 3.3	Materials: Specific to individual parts				1/1

Annex I: Self-climbing alternatives

Vehicle Definition analysed multiple alternatives to a self-climbing A/C to get to the desired altitude:

1. Carrying additional batteries to provide the power deficit during low altitudes and jettisoning them sufficient height for self-climb was reached. The extra weight entailed by the batteries would be the limiting factor, as well as the change in CG after jettisoning. However, due to a 10% margin allocated in the weight budget, and carefully tailored battery distribution (to allow for a static margin within the required bounds upon jettisoning) this is a feasible option.
2. Utilising an extra engine-propeller pair that would be optimised for climb in the lower levels of the atmosphere with the purpose of obtaining required thrust for climb at low altitudes. Engine would be jettisoned after reaching an altitude where the efficiency of the main engines would be sufficient to allow for a climb to 20 km and a full fuel-cell charge by the end of daylight. The fact that this engine would be conditioned by the maximum energy drain rate of the fuel cells and the maximum output of the solar array, if powered by them it would not be possible use this option.